

When Silence Pays: Mandatory Disclosure and Strategic Concealment in Housing Markets

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Abstract

Standard unraveling theory predicts that mandatory disclosure separates high-quality from low-quality products: silence signals bad news and non-disclosers should trade at-or-below the worst disclosed category. We document the opposite. Following France's 2021 enforcement of Energy Performance Certificate disclosure, properties listed without a certified score trade at a premium of 6.9 percent above disclosed low-efficiency properties for apartments and 3.4 percent for houses. The puzzle resolves because non-disclosure pools informed sellers strategically concealing unfavorable scores with uninformed sellers who genuinely lack a certified audit. Unable to distinguish between them, buyers treat silence as ambiguous rather than damning. We build and structurally estimate a model of the disclosure game, bringing it to the data through a variable unobservable from the buyer's side: private listing data from SeLoger, France's leading residential platform, matched to the ADEME registry of certified energy audits, give us direct observation of each seller's information status at listing time. We derive a closed-form expression for the non-disclosure premium that, in the partial-pooling equilibrium, collapses in theory to the structural concealment cost, invariant to market composition; we test this restriction directly and find it holds for houses but is rejected for apartments, a divergence we trace to a measurable over-identifying wedge, reported directly rather than resolved by construction. We extend the framework to endogenous renovation, deriving a local minimum incentive-compatible subsidy and showing that renovation subsidies, enforcement against hiding, and audit-infrastructure investment are complementary instruments acting on distinct margins. We conclude by showing that uniform national programmes are generically inefficient, with required cost-shares ranging from zero to the full renovation cost across French housing markets.

Keywords: Energy Performance Certificates, Mandatory Disclosure, Housing Markets, Unraveling, Information Asymmetry

JEL Code: Q48, R31, Q52, D83, R38, H23

1 Introduction

Mandatory disclosure is the standard regulatory response to asymmetric information about product quality. Yet the case for mandating disclosure is contested, because theory suggests markets may not need it. When quality is verifiable, the unraveling result of Grossman (1981) and Milgrom (1981) holds that sellers disclose voluntarily, since buyers rationally read silence as the worst possible news. All but the very worst sellers therefore disclose, and disclosure becomes effectively universal without a mandate. Whether this result describes real markets remains unsettled. It rests on strong assumptions, and it is difficult to test, because in most settings where quality is verifiable disclosure has long been compulsory.

To examine whether unraveling describes a market where quality is verifiable but silence is common, we study the French residential housing market around the July 2021 enforcement of Energy Performance Certificate (EPC) disclosure. Under a 2018 reform that took effect on July 1, 2021, a property's *diagnostic de performance énergétique* (DPE) became legally opposable.¹ A seller who advertises an energy score is now bound by it, and buyers acquired legal recourse against misstatement. The reform changed what a disclosed score means without changing the underlying housing stock or the legal consequences of remaining silent. It therefore isolates the

¹Article 179 III of *loi n°2018-1021 du 23 novembre 2018* (loi ÉLAN) amended Articles L271-4 and L134-3-1 of the *Code de la construction et de l'habitation*, with opposability effective July 1, 2021; the accompanying methodology overhaul (replacing the earlier invoice-based calculation with the current physical-characteristics-based method) was set by the *Arrêté du 31 mars 2021*. The *loi Climat et Résilience* (*loi n°2021-1104 du 22 août 2021*) is a separate, later reform; we invoke it elsewhere in the paper for the F/G rental-ban calendar it introduced, not for DPE opposability.

demand-side value of the signal, and it does so in a market where the certificate is issued and archived by a national agency rather than by the seller or the listing platform.

Applied to this setting, unraveling makes a sharp prediction. Once the DPE is credible, sellers of energy-efficient properties should publish their score to separate themselves; sellers of mediocre properties should then publish to avoid being pooled with worse ones; and the process should continue until listings without a score are confined to the very bottom of the quality distribution, priced at or below the worst disclosed grade. French housing is a suitable testbed for this prediction because the disclosed attribute is a single certified index with a clear mapping into a running cost buyers face directly, because the certificate is verifiable against an independent public registry, and because a large stock of properties entered the post-reform market without one.

The unraveling prediction rests on four assumptions: (a) buyers know sellers possess private information about quality, (b) disclosure is costless, (c) disclosure is credible, and (d) buyers are homogeneous in how much they value the disclosed attribute. Three of the four are either directly satisfied in this setting or can be dispensed with. Assumption (c) holds by construction after July 2021, since the DPE is issued by a certified third-party diagnostician, archived in the registry of the *Agence de la Transition Écologique* (ADEME), and legally binding on the seller. Assumption (b) is close to satisfied, because a DPE is a one-off outlay of roughly 100–250 € according to ADEME's own estimates, negligible relative to residential transaction values, and it is the seller's obligation rather than the buyer's. Assumption (a), which the theoretical literature treats as a maintained hypothesis, we do not need to assume at all. By matching listings to the ADEME registry we observe directly which sellers hold a certificate at the moment of listing, and among the apartment listings we can match at the address level, 48.8% hold none. Silence in this market is therefore not always a choice, a possibility first identified by Dye (1985) and formalised by Jung and Kwon (1988), and here it is measured rather than inferred.

The descriptiveness of unraveling in this market accordingly hinges on assumption (d). Unlike disclosure costs and disclosure credibility, which a large literature has examined, the homogeneity of buyer valuations has received little attention, even though it sets the size of the disclosure incentive. A seller's penalty for withholding is proportional to the share of buyers who price the withheld attribute. Housing makes the assumption particularly implausible. A buyer of a rarely-occupied second home in a mild climate may rationally ignore an energy score that a buyer of a primary residence in a cold region prices directly, as may an investor facing the impending rental prohibition on the worst-rated properties. Evidence across many regulated markets shows that disclosure interventions fail precisely when recipients are inattentive or heterogeneous in their response (Loewenstein et al., 2014; Ben-Shahar and Schneider, 2011), and Fishman and Hagerty (2003) show formally that mandated disclosure can be ineffective, or welfare-reducing, when buyers differ in their capacity to use the disclosed information. We therefore let the share of buyers who price the EPC vary across local markets and estimate it.

Testing a theory of strategic concealment requires observing both sides of the information asymmetry at once, the seller's true information, the seller's disclosure choice and the market's response. Most empirical work observes only one or two. Studies of information asymmetry in housing infer agent advantages from equilibrium price deviations (Levitt and Syverson, 2008); analyses of certification intermediaries characterise incentives without observing individual information endowments (Lizzeri, 1999); and work on the energy-efficiency gap proxies for seller awareness through indirect variation in energy prices (Myers, 2020). We observe all three. Listing records from SeLoger, a major French residential platform, give the listing price, property characteristics, the seller's disclosure decision, and, when a score is displayed, the letter itself. The registry of the *Agence de la Transition Écologique* (ADEME) is the official national database of certified DPE audits, reported by accredited third-party assessors and compiled independently of the listing market. We match listings to the registry on address, property type, area, and floor, which identifies whether a certified audit existed at the time of listing, the one thing buyers can never observe. Because the match requires a geocoded address, the matched sample is the subset of listings that can be located at the address level rather than the full listing universe. Within that matched sample, every listing falls into one of four groups, namely truthful disclosers, strategic hidiers who hold a certificate and omit it, genuinely uninformed sellers for whom no certificate exists, and a small residual displaying a score without a verified audit. The information structure of the disclosure game is therefore observed rather than inferred, a vantage point that to our knowledge is unique in this literature. Transaction records enter separately and are not matched to individual listings. We use them to estimate the mapping from listing to realised sale price, which supports treating the listing price as the relevant outcome, since the negotiation margin in French residential sales is small and most properties transact at or near list price (Merlo and Ortalo-Magné, 2004; Genesove and Mayer,

2001; Friggit, 2013).

The data yield four descriptive findings, drawn from two distinct samples that we keep separate throughout. First, silence commands a premium rather than a penalty. Estimated on the panel of market-level hedonic price indices, and relative to the pre-reform baseline, listings without a score trade approximately 6.9% above disclosed energy-intensive properties for apartments and 3.4% for houses, the opposite of the sign unraveling predicts. Second, the premium widened abruptly at the reform rather than emerging from trend. The composition-adjusted difference-in-differences estimates evolve in parallel before July 2021, averaging +0.002 for apartments and +0.008 for houses with no discernible drift, and then shift sharply upward, stabilising around +0.044 and +0.037 by mid-2024. For apartments the level ordering already favoured non-disclosers beforehand, so what the reform changes is the magnitude of the premium rather than its sign. Third, turning to the address-matched listing sample, non-disclosure is overwhelmingly involuntary. Among apartment listings 79.8% disclose truthfully, 16.7% are genuinely uninformed, and only 3.5% hold a certificate and withhold it, with the corresponding house shares 90.4%, 8.5% and 1.1%. Fourth, the non-disclosure rate varies widely across local markets, ranging among apartments from 6.5% to 40.9% across départements, around a median of 16.5%. Together these facts are inconsistent with unraveling as a description of this market, while pointing to the mechanism that defeats it. The pool of silent sellers is dominated by the uninformed, which blunts the inference buyers can draw from silence and shelters the few who conceal deliberately.

To recover the forces behind these patterns, we build and estimate a model of the disclosure game that allows for, but does not impose, strategic concealment. Sellers differ in whether they hold a certified audit and in the quality it would reveal; buyers differ in whether they price energy performance. Informed sellers of poor properties choose between publishing a bad score and accepting the price buyers assign to silence, and that choice depends on how contaminated the silent pool is by sellers who simply have no certificate. Because the composition of the silent pool is itself the outcome of those choices, the model is closed by a fixed point in buyer beliefs, and it nests the classical benchmark. When audit coverage and buyer sensitivity are both high, the silent pool empties and the Grossman–Milgrom outcome is recovered. For intermediate values, which is where the French data sit, the model predicts partial pooling in which some informed sellers conceal and silence carries a premium.

The estimates deliver two structural magnitudes and one specification test. Identifying buyer energy-sensitivity separately from the concealment cost requires a source of variation external to prices, since the two are shown to be identified only as their product by price moments alone (Appendix B); we construct one from the capitalised energy-cost saving a sensitive buyer attaches to verified high- versus low-quality certification, built from ADEME-certified consumption bands rather than from the price data itself. The resulting share of buyers who price energy performance averages 49.5% across apartment markets and 45.8% across house markets, and the cost of withholding a certificate settles at 2.4% of property value for apartments and 3.5% for houses, on the order of a third of the notary bill a French buyer separately pays on the same transaction. Both are economically substantial magnitudes, and of the same order in two segments that trade very differently. The model predicts, in the partial-pooling equilibrium that describes the bulk of French housing markets, that the non-disclosure premium should collapse to this concealment cost exactly, invariant to market composition; we test this prediction directly rather than impose it. It holds for houses: the gap between the observed premium and the identified concealment cost is small and statistically indistinguishable from zero. It does not hold for apartments: the premium exceeds the concealment cost by a wide and precisely estimated margin, a genuine specification tension we report rather than explain away. Measurement noise from the registry match, market-level aggregation across a more heterogeneous apartment stock, an unpriced risk premium from buyer uncertainty about who is behind a silent listing, and thinner buyer bargaining positions in fast-moving urban apartment markets are all candidates we examine in Section 4.5, and none of them move the point estimates above. We maintain, rather than test, the model's prediction that informed sellers of high-quality properties never conceal; Appendix A relaxes it by reading the withheld grade from the registry and finds the structural estimates essentially unchanged.

Our paper contributes to the literature on disclosure by testing the unraveling result in a market where the informed and uninformed can be told apart. Prior work has examined the consequences of uncertainty about information endowment (Dye, 1985; Jung and Kwon, 1988), of disclosure costs, and of disclosure credibility, but has generally had to treat sellers' information as latent. By observing it directly we can decompose silence into inability and choice, and show that the first dominates the second by an order of magnitude while the second is what sustains the price premium. The finding that silence is rewarded rather than punished is not a failure of the theory so

much as a demonstration of how sensitive its central prediction is to an assumption that is rarely examined.

Our paper also provides an information-based account of the residential energy-efficiency gap (Allcott and Greenstone, 2012; Gerarden et al., 2017). The standard diagnosis appeals to split incentives, credit constraints, and inattention to future energy costs. We identify a complementary channel internal to the disclosure game. A seller holding a poor rating can renovate and publish a good one, earning the quality premium, or stay silent and earn the non-disclosure premium instead, so the premium to opacity acts as an opportunity cost of renovation and raises the private hurdle above engineering cost. This has an unexpected policy implication. Because the equilibrium absorbs enforcement through the fraction of sellers who conceal, the renovation threshold depends only on the price gap between disclosed quality levels and is independent of enforcement intensity. Renovation subsidies, enforcement against concealment, and investment in audit coverage act on three distinct margins and are complements rather than substitutes, contrary to the conventional treatment in Gillingham and Palmer (2014). We do not claim to have estimated the one true model of housing disclosure; we propose one way of reading these data through a workhorse framework, and we are explicit throughout about which parameters are point-identified and which are bounded.

The model also speaks to renovation policy. Extending the framework to an endogenous upgrade decision, we show the minimum subsidy needed to make renovation privately worthwhile reduces, in the empirically dominant partial-pooling regime, to the price gap between verified high and low quality alone, invariant to enforcement against hiding and to how many sellers are informed, which act on separate margins instead. A uniform national subsidy is accordingly inefficient. The required cost-share the state must cover ranges from zero to the full renovation cost across markets, averaging just over half in markets that need support at all, and aggregate programme costs at plausible renovation-cost benchmarks run to €9.8–33 billion, two to eight times MaPrimeRénov’s current annual budget. Efficient targeting requires conditioning subsidy design on local market composition rather than applying one national rate, and subsidies need not carry that fiscal weight alone. Enforcement against strategic hiding and investment in audit infrastructure act on separate margins, and deploying them alongside targeted subsidies reduces how much of the gap a subsidy programme has to close by itself.

Our paper draws on three strands of work. It belongs to the literature that estimates disclosure models structurally in order to assess the descriptiveness of classical disclosure theory and to recover primitives such as information endowment, disclosure costs, and beliefs; closest to us is Bourveau et al. (2025), who structurally examine whether the unraveling result describes an unregulated nineteenth-century market and find that full disclosure is reached only gradually. It is closely related to the extensive evidence on the capitalisation of energy labels in housing, which documents robust premia for efficient properties wherever labels have been studied (Brounen and Kok, 2011; Hyland et al., 2013; Fuerst et al., 2016; Frondel et al., 2018; Myers et al., 2022); that literature establishes that labels are priced, whereas our question is why their *absence* is priced favourably. And it builds on work on information asymmetry in real-estate markets, where the informational advantage of one side of the transaction has been inferred from prices and time-on-market (Levitt and Syverson, 2008; Lizzeri, 1999).

The features that make this setting informative also bound it. French EPC disclosure is a single mandate, in one country, covering one verifiable attribute, and the low audit coverage that lets us separate inability from choice is itself a feature of a young disclosure regime. The mechanism, however, is not specific to energy labels. Wherever a disclosure mandate is introduced before the infrastructure that makes disclosure feasible is universal, a pool of genuinely uninformed non-disclosers will exist, and that pool will shelter strategic concealment and dull the incentive to disclose. That configuration characterises a broad class of emerging disclosure regimes, from climate-risk and emissions reporting to supply-chain and sustainability disclosure, in which mandates routinely precede measurement capacity, including the French carbon-disclosure mandate studied by Bourveau et al. (2024), where the same gap between obligation and measurement infrastructure arises.

2 Unraveling with Possibly Uninformed Sellers

2.1 The Unraveling Benchmark

Asymmetric information about product quality can prevent trade from happening at the right prices. If buyers cannot verify quality before purchasing, they bid no more than the value of what they expect to receive on average. Sellers of genuinely good products are then underpaid, and may withdraw from the market altogether, leaving buyers facing a pool of worse quality than they had hoped for (Akerlof, 1970). The outcome is bad for both sides.

Buyers pay for uncertainty they would rather not bear, and the best sellers cannot capture the value of what they own.

Disclosure appears to resolve this problem without any regulatory intervention. Suppose quality is verifiable, so that a seller who makes a claim about it can be held to that claim. A seller of the highest-quality product will disclose, because staying silent means being valued at the average of the pool rather than at what the product is actually worth. Buyers, anticipating this, understand that any seller who remains silent is not of the highest quality, and revise downward what they will pay for silence. That revision gives the second-highest-quality seller the same incentive, and disclosing becomes worthwhile for them too. The argument repeats. Each round of disclosure worsens the expected quality of the remaining silent pool, which strengthens the incentive for the best of those who remain to separate themselves. The process continues until all sellers but possibly the very worst have disclosed. This is the unraveling result of Grossman (1981) and Milgrom (1981). Even sellers with bad news disclose, because they anticipate that silence will be read as worse news still. If it holds, market forces alone deliver near-complete disclosure and a mandate is redundant.

The argument is powerful but it is not free. It rests on four assumptions. First, (a) buyers know that sellers possess private information about quality; if a seller may simply not know, silence no longer implies unwillingness to reveal. Second, (b) disclosure is costless, or at least cheap relative to the price consequences of silence; a large literature examines how disclosure costs truncate the unraveling process and leave a residual pool of non-disclosers (Verrecchia, 1983; Jovanovic, 1982). Third, (c) disclosure is credible, so that a claim about quality cannot be fabricated; without verifiability, disclosure degenerates into cheap talk and the unraveling logic never starts (Crawford and Sobel, 1982; Stocken, 2000). Fourth, (d) buyers are homogeneous in how much they value the disclosed attribute, so that the price penalty for silence is felt by every seller with equal force.

The first three assumptions have received sustained attention. The fourth has not, even though it governs the strength of the entire mechanism. What drives a seller to disclose is the revenue lost when buyers price silence pessimistically, and that loss is proportional to the share of buyers who care about the attribute in the first place. Whether unraveling describes any particular market is therefore an empirical question, and answering it requires knowing which of these assumptions fails there.

2.2 Two Departures and Why They Compound

Two of the four assumptions are implausible in markets where quality is certified by a third party and buyers differ in their exposure to the certified attribute.

The first departure concerns assumption (a). When some sellers genuinely lack private information, silence pools those who *will not* disclose with those who *cannot*. Dye (1985) showed that this possibility breaks the buyer inference on which unraveling depends, because a rational buyer confronted with silence must form a posterior over two distinct populations, and how pessimistic that posterior should be depends on their relative masses rather than on the act of silence itself. Jung and Kwon (1988) characterised the resulting equilibria and showed that partial pooling, in which some informed sellers of poor quality withhold while the uninformed are silent by necessity, is generically stable. Disclosure does not unravel to the bottom; it stops somewhere in the interior.

The second departure concerns assumption (d). If only a fraction of buyers price the disclosed attribute, a seller who withholds it faces a diluted penalty, because part of the demand side is indifferent to what was withheld. Fishman and Hagerty (2003) show formally that mandated disclosure can be ineffective, and even welfare-reducing, when buyers differ in their capacity to use disclosed information. The disclosure incentive is thus scaled by the composition of demand, and where that composition is unfavourable the incentive can be too weak to sustain separation even among sellers who are perfectly well informed.

The two departures do not merely coexist; they compound. A larger mass of sellers who cannot disclose dilutes the adverse inference buyers draw from silence, which raises the expected quality of the silent pool and makes withholding more attractive to informed sellers of poor quality. Those sellers then join the pool, and while their entry worsens its true average quality, buyers cannot separate them from the involuntarily silent. Meanwhile, a demand side in which many buyers do not price the attribute weakens the penalty that would otherwise discipline them. Put together, the consequence is stronger than persistence of non-disclosure. Silence can command a *premium* over the worst disclosed quality, since the silent pool contains a substantial mass of sellers whose silence carries no adverse information at all. The sellers who cannot disclose do not intend to shelter anyone.

Their presence is involuntary. But it is precisely what makes silence ambiguous, and ambiguity is what makes concealment privately rational.

Because the two departures operate on different primitives, they map out a space of possible market outcomes rather than a single prediction. Figure 1 organises that space along two dimensions, the share of sellers who hold verifiable information and the share of buyers who price it. Full unraveling requires both to be high. Where either is interior, silence survives in equilibrium, and where the informed share is low while buyer sensitivity is not, silence can be rewarded rather than punished.

	Fully informed sellers ($\pi = 1$) GM dep. 2	Partially informed sellers ($\pi < 1$)
Buyers do not care ($\rho = 0$)	Disclosure Irrelevant No demand response to disclosure Premium: $\Delta = 0$	Disclosure Irrelevant No demand response to disclosure Premium: $\Delta = 0$
Buyers care ($\rho = 1$) GM dep. 2	Full Unraveling Grossman–Milgrom Silence is fully penalised	Sustained Premium Both channels active ($0 < \rho \leq 1$) $\Delta > 0$

Figure 1. Equilibrium outcomes under seller information and buyer sensitivity heterogeneity

Two features of this taxonomy matter for what follows. It implies that the informational content of silence is a market-level object rather than a universal one, so that two markets facing the same disclosure mandate can sit in different cells and sustain opposite price patterns. And it implies that the two primitives are separately movable by policy. Investment in certification infrastructure raises the share of sellers holding verifiable information, while the share of buyers who price the attribute responds to the legal weight the certificate carries and to the cost of the attribute itself. Which cell a given market occupies, however, cannot be settled by theory. It requires measuring the composition of the silent pool and the sensitivity of buyers separately, which is what the data described in the next section allow us to do.

3 The French Energy-Performance Disclosure Regime

This section sets out the institution, the data, and four features of the market that any candidate theory of EPC disclosure has to accommodate. We lead with apartments throughout and treat houses as a corroborating segment. The two display the same qualitative patterns, but the apartment market has the lower audit coverage, the larger share of strategic concealment, and the stability across the reform that the identification strategy exploits.

3.1 Institutional Background

The French *diagnostic de performance énergétique* (DPE) is a standardised energy audit of a dwelling, carried out by an accredited third-party assessor and recorded in a national registry maintained by the *Agence de la Transition Écologique* (ADEME). It assigns a letter grade from A to G summarising modelled energy consumption per square metre. Two features make it unusual as a disclosure instrument. It is certified rather than self-reported, so a displayed grade can be checked against the registry. And the registry is centralised and public, so the *existence* of an audit is verifiable independently of both the seller and the listing platform. That second property is what makes the empirical exercise in this paper possible.

Before July 2021, the French DPE was a paper exercise. Scores were methodologically inconsistent across assessors, carried no legal weight, and were routinely ignored in negotiations: a buyer who later discovered a disclosed rating was misleading had no legal recourse. The reform of July 1, 2021 changed the regime fundamentally. It standardized the methodology, imposed stricter accreditation requirements on assessors, and, most importantly,

rendered disclosed EPC ratings legally opposable: a buyer can now use a disclosed rating as grounds for price renegotiation or legal action. The DPE became, for the first time, a document with contractual force.

Two things the reform did not change matter as much as what it did. First, the obligation to commission a DPE before marketing a property predates it. The requirement, and the liability attached to marketing a property without the required diagnostic, were established by *loi ALUR* (Article L271-4 of the *Code de la construction et de l'habitation*) well before 2021. The legal exposure a seller incurs by coming to market without a certificate is therefore governed by a mandate that straddles the reform date unchanged, while what July 2021 altered is how much a *disclosed* grade is worth to the buyer. Second, the reform sharpened a pre-existing regulatory threshold rather than inventing one. Under *loi Climat et Résilience*, dwellings rated E, F, or G face a phased prohibition on new lettings. This draws a bright line between the bottom of the grade distribution and the rest, and gives the binary quality distinction used below a concrete regulatory anchor rather than an arbitrary cutoff.

3.2 Data and Measurement

Testing a theory of strategic concealment requires observing something buyers cannot, namely whether a silent seller holds a certificate. We recover it by matching listings against the ADEME registry.

3.2.1 Sources and Sample Construction

The empirical analysis draws on two sources matched at the listing level, together with transaction records used separately. Listing records are sourced from SeLoger, a major French residential real-estate platform covering sales across metropolitan France. For each listing, we observe the listing price, property characteristics (X_i : area, number of rooms, floor, construction era, location), the disclosure indicator $d \in \{0, 1\}$, and, when $d = 1$, the displayed EPC letter (A through G). Transaction records are not matched to individual listings; we use them to estimate the mapping from listing price to realized sale price (γ_0 and γ_1 in equation (Eq. 1)), which supports treating the listing price as the relevant outcome for the disclosure decision. The ADEME registry is the official national database of certified Diagnostic de Performance Énergétique (DPE) audits, maintained by France's environment and energy management agency and reported by accredited third-party assessors. We match listings to the ADEME registry using address, property type, area, and floor, assigning to each listing an indicator $K = 1$ if a certified audit record exists at the time of listing, and $K = 0$ otherwise. For the structural estimation, listings for which no address is available are dropped, as the ADEME match requires a geocoded address; these account for a small share of the total sample.

The full listing sample covers 2021Q3 to 2024Q4, the post-reform period during which EPC disclosure is legally binding and DPE scores are credible to buyers. The apartment sample comprises 843,621 listings and the house sample 687,878 listings. For structural estimation we restrict to 2022 to 2024: the H2 2021 period (the six months immediately following the reform) is excluded because the market had not yet fully repriced the DPE (hedonic estimates of $\hat{\beta}_1$ are near zero or negative in that window), yielding 767,000 apartment listings and 633,000 house listings for the structural sample. For 51.2% of apartment listings and 40.3% of house listings, the ADEME match yields $K = 1$; the remainder are classified as genuinely uninformed ($K = 0$, no verified audit on record). For identification of market-level price indices, we use the panel of hedonic price indices constructed in Patras (2024), covering 110 markets (attraction zones, *zones d'emploi*) with monthly observations from October 2018 through July 2024.

3.2.2 Four-Group Classification

Combining the listing disclosure decision d with the ADEME information status K generates four mutually exclusive groups, summarized in Table 1.

Three features stand out. First, strategic hiding is rare but economically meaningful: 3.5% of apartment listings and 1.1% of house listings come from informed sellers who chose not to disclose. Pooling across listings, the hiding rate among informed sellers ($K = 1$) is 6.9% for apartments and 2.7% for houses; the corresponding unweighted means across markets, reported as $\hat{\sigma}_{0,m}$ in the results, are 7.7% and 3.7%. Second, silence is uncommon relative to non-verification: only 16.7% of apartment listings and 8.6% of house listings are silent sellers with no registry record, and these are the non-disclosers who provide involuntary cover for strategic hidiers. Third, and quantitatively dominant, 32.2% of apartment listings and 51.2% of house listings display a score with

Table 1. Classification of listings and sample composition

Group	Interpretation	d	K	Apartments	Houses
Truthful disclosers	Informed seller reveals θ	1	1	47.7%	39.2%
Strategic hidiers	Informed seller conceals θ	0	1	3.5%	1.1%
Genuinely uninformed	Silent; no registry record	0	0	16.7%	8.6%
Disclosed, unmatched	Displays EPC; no post-2021 registry record	1	0	32.2%	51.2%
Total				843,621	687,878

Notes. Shares of the full listing sample of each property type, computed from the listing–registry match. The “disclosed, unmatched” group displays an EPC score for which no record appears in the public ADEME registry, which contains only certificates issued from July 2021 onward; these listings are classified as truthful disclosers under the no-falsification assumption. Sellers with no post-2021 registry record ($K = 0$) therefore account for 48.8% of apartment and 61.1% of house listings, the large majority of whom do disclose a score. Columns may not sum to exactly 100% because of rounding.

no matching registry record. Taken together, sellers without a post-2021 registry record account for 48.8% of apartment listings and 61.1% of house listings, but the large majority of them do disclose a score.

Three features of this match bear on its credibility. It is a match to an administrative registry rather than to a survey or a self-report, so the information-status indicator does not depend on seller cooperation and cannot be strategically misreported. It requires a geocoded address, so the matched sample is the subset of listings that can be located at the address level rather than the full listing universe; sample scopes are stated alongside the statistics reported below. And, most consequentially, $K = 0$ must be read as *no post-July-2021 certificate in the public registry* rather than as *no certificate exists*. The registry was populated from the reform date onward, while a certificate issued earlier could remain legally valid at the time of listing, so a seller could hold and display a valid certificate that the registry cannot corroborate. The data bear this out. The share of listings displaying a score with no registry record falls monotonically as the registry accumulates, from 62.5% of apartment listings in 2021Q3 to 18.4% in 2024Q4, mirrored by a rise in verified disclosers from 7.4% to roughly one half.

This coverage gap has a known direction. A seller holding a still-valid pre-reform certificate who withholds it is recorded as genuinely uninformed rather than as a strategic hider, so the measured hiding rate $\hat{\sigma}_{0,m}$ understates the true one. Because the equilibrium belief $e(\sigma_0)$ is decreasing in σ_0 , an understated hiding rate implies an overstated belief and hence an overstated concealment cost. The coverage gap therefore reinforces, rather than threatens, the interpretation of $\hat{c}_{0,m}$ as an upper bound. It is also one reason the structural sample begins in 2022, since the second half of 2021, where the gap is most severe, is excluded.

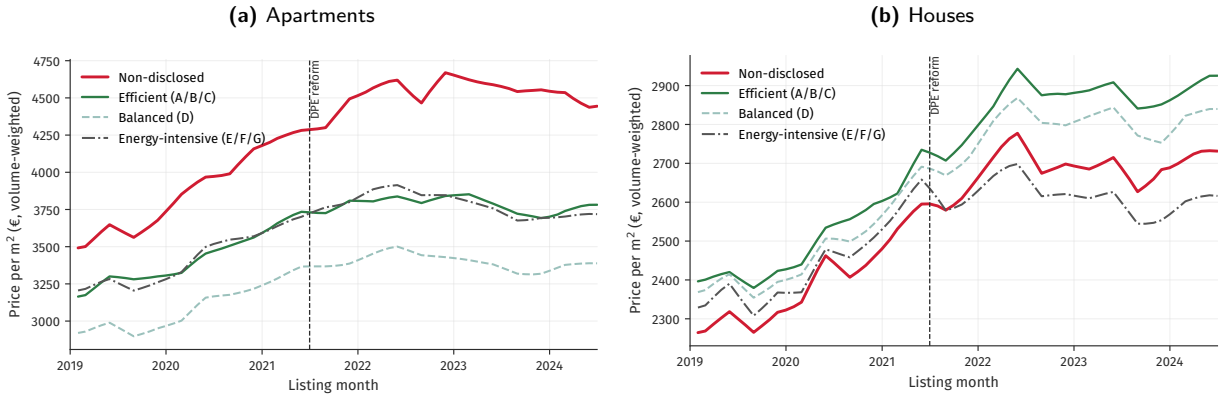
3.3 Stylized Facts

Four features of the market discipline any candidate theory. Each is documented for both segments, apartments first.

3.3.1 Non-Disclosing Listings Command a Price Premium, Widened by the 2021 Reform

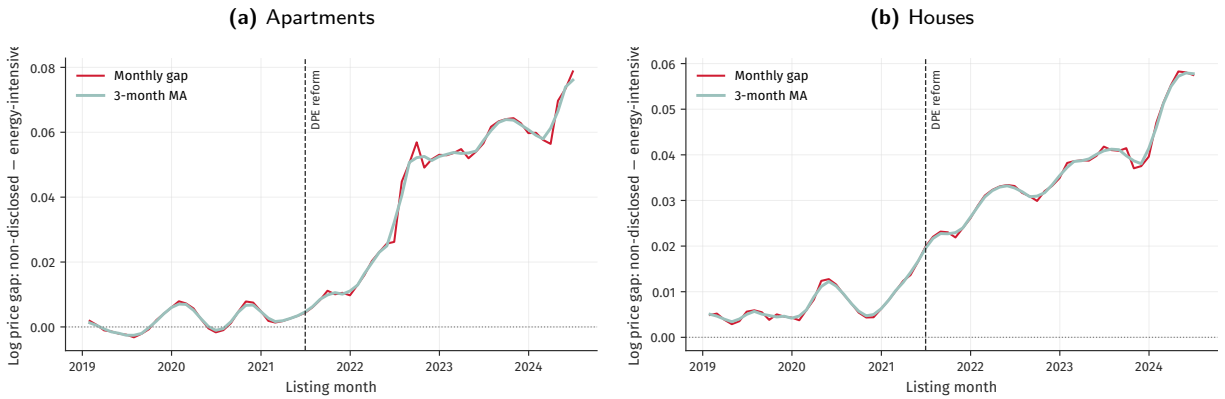
The starting puzzle is that silence about energy performance commands a price premium relative to the worst certified scores in both property types. Figure 2 plots the volume-weighted average of market-level hedonic price indices per square metre for each EPC disclosure category from January 2019 to June 2024, where each index holds property characteristics constant within a local market so that comparisons across disclosure categories reflect differences in the disclosure decision rather than differences in property type or location. For apartments (panel (a)), non-disclosing listings occupy the highest price tier across the *entire* sample window: the ordering Non-disclosed > Efficient (A/B/C) > Energy-intensive (E/F/G) is visible as early as 2019, well before the reform, and the gap widens sharply after July 2021. For houses (panel (b)), the pre-reform picture differs: non-disclosers begin near the bottom of the price distribution and rise toward the middle post-reform, reflecting the higher baseline audit coverage in the house segment documented in Section 3.3.3. The pre-reform level premium for apartments is consistent with the concentration of non-audited stock in high-price urban markets. Figure 3 isolates the *change* in the gap: it plots difference-in-differences estimates of the hedonic-controlled log price gap between

Figure 2. Price per m² by EPC disclosure category



Notes. Volume-weighted national average of market-level hedonic price indices per square metre (€) by EPC disclosure category, January 2019 to June 2024. Price indices are constructed in Patras (2024) using hedonic regressions estimated separately within each *zone d'emploi* (INSEE clusters of municipalities), covering 110 local markets. National series are obtained by aggregating market-level indices by listing volume. Series are smoothed with a centred three-month moving average. The vertical dashed line marks the July 2021 DPE reform. Source: Patras (2024); authors' calculations.

Figure 3. Non-disclosure price premium around the July 2021 DPE reform



Notes. Monthly volume-weighted log gap between the price index of non-disclosing listings and certified energy-intensive (E/F/G) listings, January 2019 to June 2024. The vertical dashed line marks the legal opposability of the DPE (July 2021). The thick line is a centred three-month moving average. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

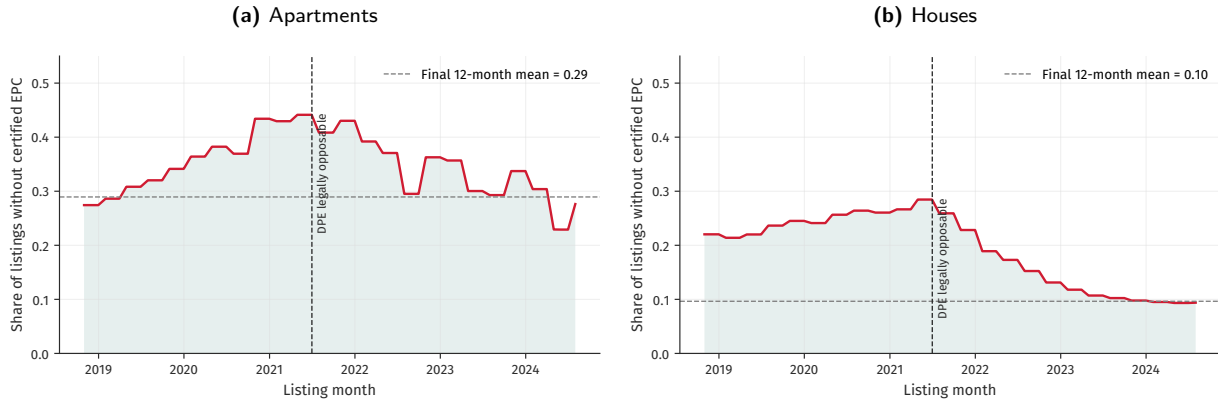
non-disclosing and energy-intensive listings, purging observable composition differences and measured relative to the pre-reform baseline. Because the estimates are normalised to that baseline, the pre-reform coefficients are informative about trends rather than levels. They average +0.002 log points for apartments and +0.008 for houses and display no discernible drift, indicating that the two series evolved in parallel before July 2021. This is precisely the identifying assumption the design requires; it is not a claim that non-disclosing and energy-intensive listings were priced alike, and Figure 2 shows that for apartments they were not. After July 2021, when DPE certificates became legally opposable under *loi Climat et Résilience*, the estimates shift abruptly upward and stabilize around +0.044 log points for apartments and +0.037 log points for houses by mid-2024. The reform therefore does not create the level difference so much as widen it sharply, in both the raw series and the composition-adjusted estimates. Classical unraveling predicts the opposite sign: silence should signal bad news and trade *below* verified low-quality listings.

3.3.2 Non-Disclosure Persists Long After Mandatory Disclosure

Classical unraveling further predicts that, once disclosure is mandatory and verifiable, the share of non-disclosing listings should converge rapidly toward zero. The French data reject this prediction in both segments, though

with different magnitudes. Figure 4 plots the monthly share of listings without a certified EPC from October 2018 through July 2024. For apartments (panel (a)), the share averages 35.9% before the reform and 33.8% afterward; over the final twelve months of the sample, it stabilizes at 28.9%. For houses (panel (b)), the corresponding numbers are 24.6% pre-reform, 14.4% post-reform, and a final-twelve-month mean of 9.7%. Houses thus display better baseline audit coverage and faster post-reform convergence, but in both segments the share remains structurally bounded away from zero more than three years after disclosure became legally required. Almost three-tenths of apartment listings and one-tenth of house listings still come to market without a certified audit, indicating that a substantial fraction of sellers cannot disclose, because they do not possess a certified audit at the time of listing.

Figure 4. Persistence of non-disclosure after the DPE reform



Notes. Monthly share of listings without a certified EPC, October 2018 to July 2024. The horizontal dashed line marks each segment's final-twelve-month mean (0.29 apartments, 0.10 houses). The vertical dashed line marks the July 2021 reform date. Volumes are summed across markets within each segment. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

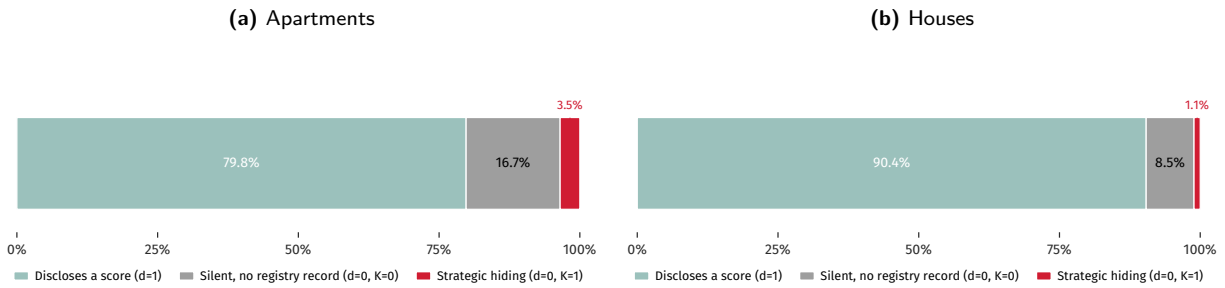
3.3.3 Direct Decomposition of the Listing Pool by Information Status

A central methodological contribution of this paper is that, by matching every listing against the ADEME registry, we can directly observe whether a seller is informed ($K = 1$, audit exists) or uninformed ($K = 0$, no audit on record). Crossing this latent information status with the seller's disclosure choice $d \in \{0, 1\}$ yields a four-cell classification of which only three cells are economically distinct: truthful disclosers ($K = 1, d = 1$), strategically hiding sellers ($K = 1, d = 0$), and genuinely uninformed sellers ($K = 0, d = 0$).² Figure 5 reports the resulting decomposition for both segments. In apartments (panel (a)), 79.8% of listings are truthful disclosers, 16.7% are genuinely uninformed, and 3.5% are strategically hiding. In houses (panel (b)), the corresponding shares are 90.4%, 8.5%, and 1.1%. Houses systematically display higher baseline disclosure and a smaller strategic-hiding share, but the qualitative structure is identical: a non-trivial mass of strategic hidiers coexists with a much larger mass of genuinely uninformed sellers in both segments.

Two features of this decomposition are central to the analysis. First, the non-disclosing pool ($d = 0$) is mechanically composite in both segments: roughly five-sixths of apartment non-disclosers and seven-eighths of house non-disclosers are genuinely uninformed, while the remainder are strategically concealing. The vast majority of silent listings are silent by constraint, not by choice. Second, the relative weight of these two subgroups varies

²The fourth cell ($K = 0, d = 1$) comprises sellers who display an EPC score for which no matching record appears in the public ADEME registry. These listings are treated as truthful disclosers, consistent with the no-falsification assumption: a score published by the owner is taken at face value, and only omission is treated as a strategic act. Two institutional features explain the size of this group. First, the public registry contains only certificates issued from July 2021 onward, whereas a certificate issued earlier could still be legally valid at the time of listing. Under *décret* n° 2020-1610 of 17 December 2020, certificates issued between 1 January 2013 and 31 December 2017 remained valid until 31 December 2022, and those issued between 1 January 2018 and 30 June 2021 until 31 December 2024, while certificates issued from 1 July 2021 are valid for ten years. A seller listing during our sample window could therefore display a still-valid certificate that is absent from the registry by construction. Second, the model assumes no falsification. Consistent with the first explanation, the share of listings in this cell falls monotonically as the registry accumulates post-reform certificates: from 62.5% of apartment listings in 2021Q3 to 18.4% in 2024Q4, and from 73.1% to 43.7% for houses.

Figure 5. Three-group decomposition of the listing pool



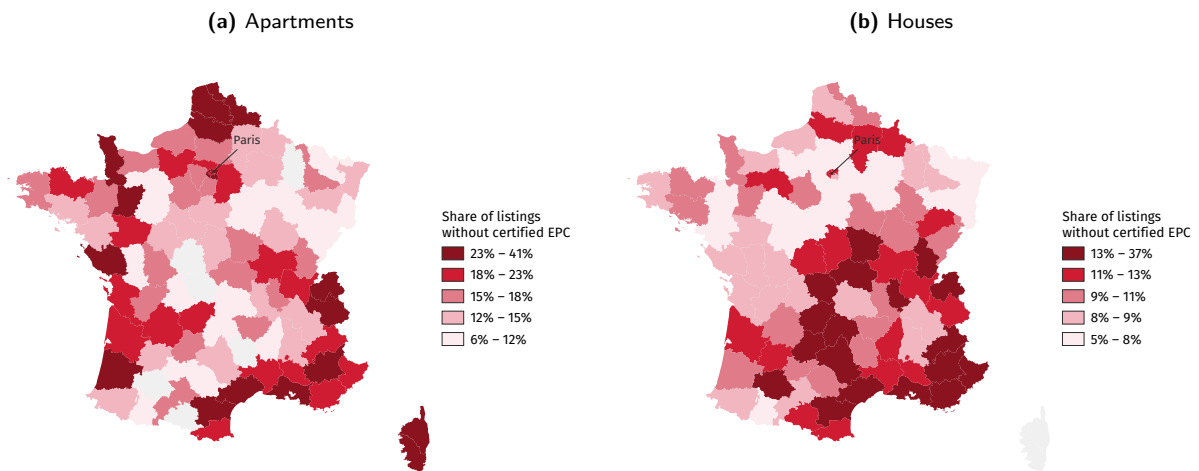
Notes. Shares from the estimation sample of each property type: 834,816 apartment listings and 677,011 house listings, after registry-match cleaning and market-size filtering. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

substantially across property types and, as we show below, across local markets, with direct implications for the equilibrium informativeness of non-disclosure.

3.3.4 Substantial Cross-Market Heterogeneity in Non-Disclosure

Figure 6 maps the apartment (panel (a)) and house (panel (b)) non-disclosure rates at the département level. For apartments, the rate ranges from 6.5% in Aube to 40.9% in Corse-du-Sud, with a cross-département median of 16.5%; high values cluster in southern France, Corsica, and parts of the Parisian region (Seine-Saint-Denis at 32.1%). For houses, the range is 5.4% in Essonne to 37.4% in Paris, with a median of 10.0%; the Paris département is an outlier on the high end, while southern and rural départements lie closer to the metropolitan baseline. The geographic patterns differ across the two segments, reflecting the different local composition of each property type's listing pool, but the underlying message is identical: within-segment cross-département dispersion is large in both cases. This heterogeneity rules out a national-level treatment of the information environment in either segment: any credible structural exercise must allow the depth of audit coverage and the salience of EPC information to vary by local market.

Figure 6. Non-disclosure rate by département, July 2021 to December 2024



Notes. Shares are computed on listings with a valid département code; départements with fewer than 200 listings within each segment are shown in light grey, and overseas départements are omitted. Classification: quantiles ($k = 5$) on metropolitan France, computed separately within each panel so the breakpoints reflect each segment's distribution. Projection: Lambert-93 (EPSG:2154). Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; département boundaries from france-geojson (G. David, 2024).

3.4 What the Facts Demand of the Theory

Taken together, the four stylized facts impose tight discipline on any candidate theory of EPC disclosure. The documented non-disclosure premium rules out classical full-unraveling models with skeptical buyers, in which silence must trade at or below the worst certified score. The persistence of non-disclosure after the reform rules out frictionless implementations in which enforcement is complete and disclosure is universal at any positive cost of concealment. The composite nature of the non-disclosing pool requires the theory to recognize that silence aggregates two economically distinct subpopulations: informed sellers who choose to conceal unfavorable information, and uninformed sellers who cannot disclose because no certified audit exists. Their joint mass determines the informational content of non-disclosure and cannot be collapsed into a single type. Finally, the substantial cross-market heterogeneity in non-disclosure rates and price gaps rules out a uniform national treatment of these structural parameters: estimation must be conducted market by market. The model developed in the next section is designed to satisfy these four requirements simultaneously.

4 A Structural Interpretation

4.1 Motivation

To recover the forces behind the patterns documented in Section 3, we specify and estimate a model of the seller's disclosure decision. The model is a housing-market application of Dye (1985), as equilibrium-characterised by Jung and Kwon (1988), with one structural addition, heterogeneity in buyer EPC sensitivity (ρ). The binary quality space ($\theta \in \{0, 1\}$) and the seller-information heterogeneity (π) are adapted directly from those antecedents. What goes beyond them is the closed-form expression for the non-disclosure premium and its invariance to market composition in the partial-pooling regime (Proposition 4.2), the strict cross-regime ordering of the premium (Eq. 7), and an identification strategy that uses the directly observed audit status K to recover c_0 and ρ market by market.

We allow for, but do not impose, strategic concealment. Nothing in the specification requires informed sellers to withhold. The hiding rate is an equilibrium object, the model admits full unravelling as a special case, and whether concealment occurs, and how much, is what the estimation determines.

Two considerations motivate this structure. The first is resemblance. The facts in Section 3.3 line up with the Dye–Jung–Kwon configuration, in which a silent pool is overwhelmingly silent by constraint rather than by choice, a small but non-trivial mass of informed sellers who withhold, and a premium attached to silence rather than a penalty. The second is institutional. The binary quality space has an empirical anchor in the rental-ban threshold introduced by *loi Climat et Résilience*: dwellings rated E, F, or G face a progressive prohibition on new lettings, which draws a bright line between the bottom of the grade distribution and the rest and maps onto $\theta \in \{0, 1\}$ without an arbitrary cutoff.

A competing explanation must also clear a demanding timing test. Any account that attributes the post-reform widening to a supply-side shift has to generate an abrupt change at July 2021 in *both* the quality premium and the non-disclosure premium, in the same markets and on the same calendar date, while leaving the composition of the listing pool undisturbed. The share of listings without a disclosed score is, however, essentially flat across the reform in the apartment segment (Section 3.3.2), which is the segment where identification is sharpest. A shift in the mix of sellers would generically move the two gaps by different amounts, and would show up in that share. Neither is observed.

The mechanism itself is the one set out in Section 2: sellers who cannot disclose dilute the adverse inference buyers draw from silence, which makes withholding attractive to informed sellers of poor quality, whose entry the buyer cannot separate from the involuntarily silent. France sits in the bottom-right cell of Figure 1: incomplete registry coverage combined with positive buyer EPC-sensitivity. Even where the EPC-sensitive buyer share approaches one, as it does in most apartment markets after the reform, incomplete coverage alone sustains the premium, because so long as a non-trivial share of non-disclosers hold no certificate, buyers cannot fully penalise silence. Full unravelling requires both conditions; France satisfies only one.

Finally, we are explicit about what the model does not settle. It takes no stance on *why* an informed seller withholds, since deliberate concealment and an unrecorded but still-valid certificate are observationally similar in the listing data, and we do not attempt to separate them. It assumes no falsification, so a displayed score is taken at face value. And it maintains Corollary 4.1, $\sigma_1^* = 0$, rather than testing it. Informed non-disclosers are assigned

low quality by revealed preference, which is an assumption and not a result. Appendix A reports estimates that instead read the withheld quality from the registry, so the restriction can be assessed. Nor does the design pin down ρ_m from the interior in most markets, where the data are consistent with full buyer sensitivity but cannot establish how close to one it lies, and we maintain $\hat{\rho}_m = 1$ for those markets and treat $\hat{c}_{0,m}$ as an upper bound accordingly. We do not purport to estimate the one true model of housing disclosure; we propose one way of reading these data through a workhorse framework.

4.2 Setup

4.2.1 Environment and Timing

A seller owns a property with observable attributes X and an unobservable energy performance level $\theta \in \{0, 1\}$, where $\theta = 1$ denotes high efficiency. Whether the seller knows θ depends on whether she has obtained a certified Diagnostic de Performance Énergétique (DPE) audit prior to listing. With probability $\pi \in (0, 1)$, the seller is *informed* ($K = 1$) and knows θ ; with probability $1 - \pi$, she is *uninformed* ($K = 0$) and does not. The parameter π is not a choice variable: it reflects structural features of local audit markets, including the cost and availability of accredited assessors, the time pressure sellers face (relocation, financial distress, inheritance), and the regulatory requirements in force. It is measured directly from the ADEME registry match, as described in Section 3.2.

The seller simultaneously chooses a disclosure decision $d \in \{0, 1\}$ and a listing price p^{list} . French property law requires that all mandatory diagnostic documents, including the DPE, be appended to the sales agreement at signing (loi ALUR, Article L271-4 CCH), so the disclosure decision and the listing price are set jointly at the time the property is placed on the market. This institutional timing motivates the simultaneous-choice formulation: the seller does not first decide whether to disclose and then price, but sets a complete strategy that internalizes how the disclosure decision shifts buyer beliefs and thereby affects both the probability of sale and the expected transaction price.

Assumption (No Falsification). *Disclosed EPC scores are verifiable against the ADEME registry. An informed seller with $K = 1$ may choose $d \in \{0, 1\}$, while an uninformed seller with $K = 0$ is constrained to $d = 0$: she cannot disclose what she does not possess. This rules out costless misreporting.*

The key friction is that buyers observe (d, p^{list}, X) but not the seller's information status K : they cannot distinguish a seller who chose not to disclose from one who could not. A buyer then arrives and decides whether to purchase at p^{list} . With probability $\rho \in (0, 1)$, the buyer is *EPC-sensitive* and incorporates energy performance into her valuation; with probability $1 - \rho$, the buyer is *EPC-insensitive* and conditions her bid solely on observable attributes. Sellers cannot observe buyer type ex ante and optimize against expected demand. The parameter ρ is market-specific, reflecting the local composition of the buyer pool, and is identified from the differential price response to the July 2021 reform across markets, as detailed in Section 4.4.

We provide a visual representation of the model dynamics and timing in Figure 7; a fuller schematic overview appears later in Figure 14.

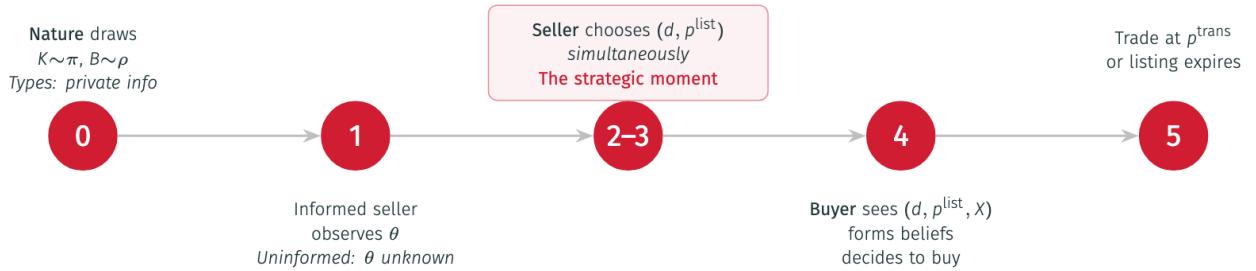


Figure 7. Model timing

4.2.2 Payoffs and Hiding Costs

In French residential markets, listing prices anchor negotiation rather than serving as posted take-it-or-leave-it offers, and the transaction price conditional on a sale typically falls below the listing price by a margin that depends

on local market conditions and property characteristics.³ Buyer arrival and match quality are stochastic, so the seller faces uncertainty over both whether a sale occurs and, conditional on a sale, at what price. Her problem therefore separates into a transaction probability Γ and an expected transaction price ϕ , each depending on the listing price and disclosure decision:

$$\max_{\{d, p^{\text{list}}\}} \Pi = \underbrace{\mathbb{E}[p^{\text{trans}} \mid d, p^{\text{list}}, X]}_{\phi(p^{\text{list}})} \cdot \underbrace{\Pr(T=1 \mid d, p^{\text{list}}, X)}_{\Gamma(p^{\text{list}})} - C(\theta, d) \quad (\text{Eq. 1})$$

where $\phi(p^{\text{list}}) = \gamma_0 + \gamma_1 \cdot p^{\text{list}} + \sum_n \gamma_n X_n$ is the expected transaction price conditional on a sale occurring, and $C(\theta, d) = \mathbf{1}[d=0] \cdot (c_0 + \theta c_1)$ is the hiding cost, paid only when the seller withholds information.

The hiding cost has two components. The term $c_0 \geq 0$ applies to all non-disclosers, regardless of quality. In the French institutional context it captures the regulatory burden imposed by loi ALUR, from the legal exposure if a buyer later discovers a concealed audit via the public ADEME registry or a notaire liability check, to the informational ambiguity that prolongs negotiation and raises the risk that a sale falls through. The term $c_1 \geq 0$ is an additional opportunity cost borne only by high-quality sellers ($\theta = 1$) who choose to hide, since by not disclosing they forgo the revenue premium that verified high quality would command from EPC-sensitive buyers. Because $c_1 > 0$, hiding is strictly cheaper for low-quality sellers than for high-quality ones, so the net gain from concealment is larger for $\theta = 0$ sellers at any given pool belief, and this differential is the sorting pressure that drives full separation of high-quality sellers in equilibrium (Corollary 4.1). Hiding costs are paid unconditionally on sale, reflecting compliance burdens and reputational exposure that arise at listing time rather than at closing.

c_0 serves two purposes here. It is an institutional cost of concealment, fixed within the prevailing enforcement regime, which we recover from the equilibrium price gap in Section 4.4. It is also a policy object, a margin the regulator can shift through stricter enforcement, and Section 5 studies counterfactual variation in c_0 on that basis. Estimation reads c_0 off the current regime, while the policy exercise perturbs it around that same reading.

4.2.3 Buyer Valuations and Demand

The partition of buyers into EPC-sensitive and EPC-insensitive types reflects a well-documented empirical regularity, since willingness to pay for energy efficiency varies sharply with how much of the ongoing energy cost the buyer will actually bear. Primary-residence buyers in northern France with long heating seasons, and investor-buyers who face the impending F/G rental ban under the *loi Climat et Résilience*, have strong incentives to incorporate EPC scores into their bids. By contrast, a vacation-home buyer who occupies the property only a few weeks a year, or a buyer shopping in a mild climate, has much less riding on it and may rationally discount energy performance. This heterogeneity determines ρ , the share of the local buyer pool that is EPC-sensitive.

EPC-sensitive buyers ($B = 1$) condition their valuation on energy performance:

$$v_1(\theta, d) = \begin{cases} \beta_0 + \beta_1 \cdot \theta + \sum_n \beta_n X_n & \text{if } d = 1 \text{ (verified quality),} \\ \beta_0 + \beta_1 \cdot \mathbb{E}[\theta \mid d=0] + \sum_n \beta_n \cdot X_n & \text{if } d = 0 \text{ (inferred quality).} \end{cases} \quad (\text{Eq. 2})$$

We adopt the benchmark $\alpha_k = \beta_k$, under which buyers value quality equally whether it is certified or inferred. The assumption is conservative in a precise sense. Under the alternative $\alpha_k < \beta_k$, a distrust discount on unverified quality, the non-disclosure premium would be smaller for any given pool composition, so our identified ρ is a lower bound on the true EPC-sensitive buyer share, and the premium we document understates what a fully skeptical buyer would generate. EPC-insensitive buyers ($B = 0$) value the property solely on observables, $v_0(X) = \delta_0 + \sum_n \delta_n X_n$. The aggregate transaction probability mixes the two buyer types:

$$\Gamma(p^{\text{list}}) = \rho \cdot G(v_1(d) - p^{\text{list}}) + (1 - \rho) \cdot G(v_0 - p^{\text{list}}) \quad (\text{Eq. 3})$$

where $G(\cdot)$ is the logistic CDF governing idiosyncratic taste shocks. The parameter ρ enters the demand function as a weight on the EPC-sensitive buyer valuation, so a higher ρ amplifies the revenue difference between a disclosed

³The negotiation margin in France is empirically small. The majority of properties transact at or very near list price, and the average discount rarely exceeds three to five percent. Merlo and Ortalo-Magné (2004) document similar patterns for England, showing that most residential transactions involve little or no negotiation once a listing price is set. Genesove and Mayer (2001) provide evidence that list prices act as anchors that constrain final transaction prices. For France specifically, Friggit (2013) documents that the average negotiation margin has remained narrow and declining, with a large share of properties selling without any price reduction.

high-quality listing and a non-disclosing listing and raises the stakes of the seller's disclosure decision. Empirically, ρ is a market-level structural parameter identified from the cross-market variation in the price response to the July 2021 reform, developed in Section 4.4.

4.2.4 Buyer Beliefs and the Non-Disclosing Pool

When observing a non-disclosing listing, buyers cannot determine whether the seller is strategically concealing an unfavorable audit or simply lacks a certified EPC. They form a posterior belief about expected quality using Bayes' rule. The structure of this belief depends on the composition of the non-disclosing pool, which is itself an equilibrium object.

Who Hides? Each seller is characterized by two attributes: information status $K \in \{0, 1\}$ and quality $\theta \in \{0, 1\}$. We maintain the following assumption throughout.

Assumption 4.1 (Independence) $K \perp \theta$, with $\Pr(K = 1) = \pi$ and $\Pr(\theta = 1) = \bar{\theta}$.

Independence is most plausible for transactions driven by exogenous life events (succession following a death, court-ordered liquidation, or divorce settlement), in which legal proceedings impose binding timing constraints that preclude any quality-contingent decision to obtain a certified EPC. In such cases, the seller's information status K is determined by administrative capacity and procedural deadlines rather than the energy performance of the dwelling; a seller under a judicial mandate to liquidate an estate has neither the incentive nor the discretion to condition audit acquisition on θ . In France, forced transactions of this type constitute a substantial share of the resale market, providing a natural mechanism through which $K \perp \theta$ obtains in the data.

Table 2 summarizes disclosure behavior by seller type. Uninformed sellers have no choice, since they cannot disclose information they do not possess (No Falsification Assumption). Informed sellers choose strategically, weighing the revenue gain from disclosure against the option of pooling with the non-disclosing mass. We denote by $\sigma_\theta \equiv \Pr(d = 0 \mid K = 1, \theta)$ the equilibrium hiding probability for an informed seller with quality θ .

Table 2. Disclosure Behavior by Seller Type

Type	Mass	Can Disclose?	Hides?	Hiding Mass
Informed, $\theta = 1$	$\pi\bar{\theta}$	Yes	Prob. σ_1	$\pi\bar{\theta}\sigma_1$
Informed, $\theta = 0$	$\pi(1 - \bar{\theta})$	Yes	Prob. σ_0	$\pi(1 - \bar{\theta})\sigma_0$
Uninformed, $\theta = 1$	$(1 - \pi)\bar{\theta}$	No	Always	$(1 - \pi)\bar{\theta}$
Uninformed, $\theta = 0$	$(1 - \pi)(1 - \bar{\theta})$	No	Always	$(1 - \pi)(1 - \bar{\theta})$

Notes: $\sigma_\theta \equiv \Pr(d = 0 \mid K = 1, \theta)$ denotes the hiding probability for an informed seller with quality θ . Uninformed sellers cannot disclose and hide with probability one.

The Belief Function Applying Bayes' rule to the pool composition in Table 2, the expected quality among non-disclosers is:

$$e(\sigma_0, \sigma_1) \equiv \Pr(\theta = 1 \mid d = 0) = \frac{\pi\bar{\theta}\sigma_1 + (1 - \pi)\bar{\theta}}{\pi\bar{\theta}\sigma_1 + \pi(1 - \bar{\theta})\sigma_0 + (1 - \pi)} \quad (\text{Eq. 4})$$

The numerator counts the high-quality mass in the pool, combining informed high-quality sellers who choose to hide ($\pi\bar{\theta}\sigma_1$) with uninformed sellers who happen to have high quality $((1 - \pi)\bar{\theta})$. The denominator is the total non-disclosing mass, so the belief e is a weighted average across two economically distinct populations. When uninformed sellers dominate the pool, because π or σ_0 is small, non-disclosure looks like a constraint rather than a choice and beliefs remain favorable to the seller. If instead strategic low-quality hidiers make up most of the pool, with both π and σ_0 large, the pool is contaminated and beliefs deteriorate accordingly. Unraveling is therefore incomplete, because non-disclosure is inherently ambiguous whenever $\pi < 1$, whereas in classical disclosure theory all sellers are informed and silence is unambiguous bad news.

Lemma 4.1 (Belief Monotonicity) $e(\sigma_0, \sigma_1)$ is strictly decreasing in σ_0 and strictly increasing in σ_1 .

Proof 4.1 (Proof sketch) Raising σ_0 adds more low-quality hiders to the denominator while leaving the numerator unchanged, so e falls. Raising σ_1 instead adds the same term $\pi\bar{\theta}$ to both numerator and denominator, and because $e < 1$ the numerator effect dominates, so e rises. This strict decrease in e as σ_0 rises is what makes the equilibrium in Section 4.3 unique. As more informed low-quality sellers choose to hide, buyer beliefs worsen, which attenuates the benefit of hiding and restores incentive compatibility.

4.3 Equilibrium

4.3.1 Disclosure Incentives

A seller's disclosure decision reduces to one comparison: does withholding the audit, priced off the market's pool belief e , fetch more than disclosing the true quality θ would? A low-quality seller ($\theta = 0$) gains by pooling with uninformed sellers and capturing a share of their quality premium; a high-quality seller ($\theta = 1$) loses, because pooling with low-quality hiders drags the belief e below 1. That asymmetry is the source of everything that follows.

Proposition 4.1 (Disclosure Incentives and Seller Sorting) *Under the benchmark $\alpha_k = \beta_k$, a seller of type (K, θ) strictly prefers disclosure if and only if $\theta > e(\sigma_0^*)$. In particular:*

- (i) *An informed seller with $\theta = 1$ strictly prefers disclosure for any $e(\sigma_0^*) < 1$.*
- (ii) *An informed seller with $\theta = 0$ strictly prefers non-disclosure whenever $e(\sigma_0^*) > 0$.*

Proof. See Appendix E. $\square$

Proposition 4.1 is the familiar adverse-selection asymmetry, worked out for this setting. A high-quality seller has nothing to gain from ambiguity. Disclosure reveals her type exactly; non-disclosure pools her with low-quality hiders, and since $e(\sigma_0^*) < 1$ for any non-degenerate pool, that comparison always binds. A low-quality seller faces the opposite incentive — concealment lets her price off $e(\sigma_0^*)$, which is strictly above her true quality of zero whenever any uninformed sellers are present, so the gain from pooling is strictly positive. The hiding cost $c_1 > 0$ that falls only on high-quality concealers reinforces this separation but is not what drives it under the benchmark $\alpha_k = \beta_k$: pool inference alone is enough to make high-quality sellers disclose, and c_1 only starts to matter once $\alpha_k < \beta_k$, the distrust-discount case.

Corollary 4.1 (High-Quality Sellers Always Disclose) *In any equilibrium, $\sigma_1^* = 0$.*

Proof. Follows directly from Proposition 4.1(i). $\square$

This is a partial-unraveling result. The top of the quality distribution fully separates, since an informed high-quality seller has nothing to gain from staying silent and is actively hurt by being lumped in with low-quality hiders, so she always discloses. That clears high-quality sellers out of the non-disclosing pool in any equilibrium and leaves the entire strategic question to how informed low-quality sellers behave.

With $\sigma_1^* = 0$, the belief simplifies. Writing $\sigma_0 \equiv \Pr(d = 0 \mid K = 1, \theta = 0)$ for the fraction of informed low-quality sellers who conceal, the non-disclosing pool now contains only uninformed sellers of both types plus these strategic hiders:

$$e(\sigma_0) = \frac{(1 - \pi)\bar{\theta}}{\pi(1 - \bar{\theta})\sigma_0 + (1 - \pi)} \quad (\text{Eq. 5})$$

$e(\sigma_0)$ falls monotonically in σ_0 , because every additional hider adds weight $\pi(1 - \bar{\theta})$ to the denominator without moving the numerator, diluting the pool and worsening what buyers infer from silence.

4.3.2 Equilibrium Characterization

Write $\Delta(e) \equiv \rho\beta_1 e$ for the revenue a low-quality seller gains from non-disclosure when buyers hold belief e , and define the net hiding gain $\Phi(\sigma_0)$ as that revenue gain net of the hiding cost:

$$\Phi(\sigma_0) \equiv \Delta(e(\sigma_0)) - c_0 = \frac{\rho\beta_1(1 - \pi)\bar{\theta}}{\pi(1 - \bar{\theta})\sigma_0 + (1 - \pi)} - c_0. \quad (\text{Eq. 6})$$

Φ is strictly decreasing in σ_0 — more concealment worsens the pool, which shrinks the gain from concealing further — and this monotone feedback is what pins down a unique equilibrium without any need for refinements.

The three regimes below follow from comparing Φ against zero at its two extremes: $\Phi(0) = \Delta(\bar{\theta}) - c_0$, the net gain against the cleanest possible pool, and $\Phi(1) = \Delta(e(1)) - c_0$, the net gain against the most contaminated one.

Proposition 4.2 (Equilibrium Existence and Uniqueness) *There exists a unique equilibrium, characterized as follows:*

- (i) If $c_0 > \Delta(\bar{\theta})$, the unique equilibrium is Cost-Deterred Hiding: $\sigma_0^* = 0$ and $e^* = \bar{\theta}$.
- (ii) If $c_0 < \Delta(e(1))$, the unique equilibrium is Complete Adverse Selection: $\sigma_0^* = 1$ and $e^* = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$.
- (iii) If $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$, the unique equilibrium is Partial Pooling: $\sigma_0^* \in (0, 1)$ is the unique solution to $\Delta(e(\sigma_0^*)) = c_0$.

Proof. See Appendix E. $\square$

Table 3. Equilibrium Regimes

Equilibrium	Condition	Strategy σ_0^*	Belief e^*	Premium $p_0^* - p_1^*$
Cost-Deterred Hiding	$c_0 > \Delta(\bar{\theta})$	0	$\bar{\theta}$	$\rho\beta_1\bar{\theta}$
Complete Adverse Selection	$c_0 < \Delta(e(1))$	1	$\frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}}$	$\rho\beta_1e^*$
Partial Pooling	$\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$	$(0, 1)$	Solves $\Phi(\sigma_0^*) = 0$	$\rho\beta_1e(\sigma_0^*)$

Neither corner fits what we observe. Universal concealment would need the non-disclosure premium to beat c_0 even against the dirtiest possible pool, and universal disclosure would need concealment to be unprofitable even against the cleanest. The data show something in between, with some low-quality sellers disclosing and the rest pooling with uninformed non-disclosers. Solving (Eq. 6) for the interior case $\Phi(\sigma_0^*) = 0$ then gives a closed form for the hiding rate,

$$\sigma_0^* = \frac{1}{\pi(1 - \bar{\theta})} \left[\frac{\rho\beta_1(1 - \pi)\bar{\theta}}{c_0} - (1 - \pi) \right].$$

The non-disclosure premium follows directly:

$$p_0^* - p_1^* = \begin{cases} \rho\beta_1\bar{\theta} & \text{Cost-Deterred Hiding } (\sigma_0^* = 0), \\ c_0 & \text{Partial Pooling } (\sigma_0^* \in (0, 1)), \\ \rho\beta_1\frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}} & \text{Complete Adverse Selection } (\sigma_0^* = 1), \end{cases} \quad \text{with } \rho\beta_1\frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}} < c_0 < \rho\beta_1\bar{\theta}. \quad (\text{Eq. 7})$$

All three equilibria in Proposition 4.2 come from the same comparison. The hiding gain $\Delta(e) = \rho\beta_1e$ is evaluated at the two extremes of pool contamination and set against the hiding cost c_0 . If c_0 dominates even at the cleanest pool ($\sigma_0 = 0$, $e = \bar{\theta}$), no one hides; if the gain survives even the dirtiest pool ($\sigma_0 = 1$, $e = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$), everyone hides; in between, an interior share hides and the market settles there. The premium inherits a strict ordering across the three regimes, $\rho\beta_1\bar{\theta} > c_0 > \rho\beta_1(1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$ (Cost-Deterred Hiding above, Complete Adverse Selection below, and Partial Pooling pinned at c_0 in between), and the interior value depends on neither π nor ρ .

Cost-Deterred Hiding is where the hiding cost is prohibitive even against the cleanest pool imaginable, one made up entirely of uninformed sellers drawn from the unconditional quality distribution. No informed low-quality seller finds concealment worth it, so $\sigma_0^* = 0$ and the pool stays uncontaminated at $e^* = \bar{\theta}$, and the premium $\rho\beta_1\bar{\theta}$ is just the unconditional quality belief scaled by buyers' EPC sensitivity. The deterrence result echoes Milgrom (1981) and Grossman (1981), but the mechanism differs, since their unraveling runs through a belief penalty, where silence triggers the worst possible inference, while here it runs through the cost c_0 , precisely because uninformed

sellers keep silence from ever carrying that worst-type inference. This is the only regime where the non-disclosing pool is informationally clean.

Complete Adverse Selection is the opposite corner. The hiding gain survives even the dirtiest pool the model can produce. Every informed low-quality seller hides, $\sigma_0^* = 1$, and buyer beliefs collapse to their floor, $e^* = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$, strictly below $\bar{\theta}$, taking the premium down to its lower bound with them. That floor is nonetheless strictly positive, because the uninformed sellers in the pool keep diluting the adverse inference and block the complete belief collapse that Milgrom–Grossman unraveling would otherwise imply. This is the Dye (1985) mechanism, in which a residual, non-strategic silent population keeps silence partially informative even once every strategic hider has already piled in.

The indifference condition $\Delta(e(\sigma_0^*)) = c_0$ pins down *Partial Pooling*, the interior case between the two corners. Φ is positive at $\sigma_0 = 0$ and negative at $\sigma_0 = 1$, so strict monotonicity delivers a unique such crossing, where the marginal hider is exactly indifferent. The underlying mechanism is a feedback loop. As more low-quality sellers hide, $e(\sigma_0)$ falls, the pool looks worse, and $\Delta(e)$ shrinks, until the shrinking gain reaches the hiding cost. A fraction σ_0^* hides and pools with the uninformed; the remaining $1 - \sigma_0^*$ disclose rather than keep contaminating the pool for a gain that no longer covers c_0 .

What makes this regime useful is that the equilibrium premium equals c_0 exactly, and stays there regardless of π or ρ . That invariance is just the indifference condition doing its job. The price gap between the non-disclosing pool and disclosed low quality has to equal the hiding cost, or the marginal seller would strictly prefer one action to the other. Take a rise in π . Holding σ_0^* fixed, it would mechanically dirty the pool and push e down, but equilibrium restores $\Phi = 0$ by lowering σ_0^* instead, so fewer hidings mean a cleaner pool, e rebounds, and the premium lands back on c_0 . A change in ρ is absorbed the same way. The non-disclosure premium is therefore a revealed hiding cost rather than a free function of market composition. The observable price gap between non-disclosing and disclosed-low sellers identifies c_0 directly, and this is the only regime where π , ρ , and c_0 are all jointly identified.

4.3.3 Comparative Statics

Proposition 4.3 (Comparative Statics) *The following comparative statics hold:*

- (i) *In the Complete Adverse Selection equilibrium: $\partial(p_0^* - p_1^*)/\partial\pi < 0$ and $\partial(p_0^* - p_1^*)/\partial\rho > 0$.*
- (ii) *In the Partial Pooling equilibrium: $\partial\sigma_0^*/\partial c_0 < 0$, $\partial\sigma_0^*/\partial\rho > 0$, and $\partial\sigma_0^*/\partial\pi < 0$.*

Proof. See Appendix E. $\square$

Part (i) works through two distinct channels. The effect of π runs through the *belief channel*, since a rising share of informed sellers makes non-disclosure more informative of low quality, so e^* falls and the gain from concealment dissipates. The effect of ρ runs through the *revenue channel* instead, since a larger share of EPC-sensitive buyers widens the price spread between quality levels and raises the value of concealment.

Part (ii) follows from the implicit function theorem applied to $\Phi(\sigma_0^*) = 0$. Because $\Phi'(\sigma_0^*) < 0$, $\partial\sigma_0^*/\partial c_0 < 0$ follows directly. A higher ρ raises Δ at any fixed e , which shifts Φ upward and raises the interior equilibrium, so $\partial\sigma_0^*/\partial\rho > 0$. π works through the same belief channel as in part (i). A higher π increases the contamination weight $\pi(1 - \bar{\theta})$ in the denominator of $e(\sigma_0)$, which reduces $\Delta(e(\sigma_0))$ and shifts Φ downward, so $\partial\sigma_0^*/\partial\pi < 0$. The parameters π and c_0 are therefore *policy substitutes*, each reducing strategic hiding through a distinct channel, π through beliefs and c_0 through costs, with marginal effects on σ_0^* that are both negative and quantifiable from the estimated model.

Corollary 4.2 (Effect of ρ on the non-disclosure premium) (i) *The non-disclosure premium $\Delta = p_0^* - p_1^*$ is weakly increasing in ρ : strictly so in the Cost-Deterred Hiding and Complete Adverse Selection regimes ($\partial\Delta/\partial\rho > 0$), and zero in the Partial Pooling regime ($\partial\Delta/\partial\rho = 0$, with the premium pinned at c_0).*

- (ii) *The parameter interval sustaining the Partial Pooling regime, $[\rho\beta_1 e(1), \rho\beta_1 \bar{\theta}]$, is weakly expanding in ρ : its width $\rho\beta_1[\bar{\theta} - e(1)]$ is strictly increasing in ρ whenever $\pi > 0$.*

Proof. See Appendix E. $\square$

A larger share of buyers who price the certificate raises what silence is worth, but only at the corners. Inside Partial Pooling the premium is pinned to the concealment cost, so a rise in ρ shows up as more sellers hiding rather than a wider price gap. Part (ii) adds that higher ρ widens the range of parameters over which Partial Pooling survives at all. Empirically, this predicts that within partial-pooling markets the non-disclosure premium should be uncorrelated with $\hat{\rho}_m$ even as the hiding rate rises with it, and that markets with higher $\hat{\rho}_m$ should be less likely to sit in a corner regime — a pattern we check directly against the regime classification in Section 4.5.

Proposition 4.4 (Equilibrium effect of π) *As π increases, the lower boundary $\Delta(e(1))$ of the Partial Pooling interval falls ($\partial e(1)/\partial \pi < 0$), so the condition for Complete Adverse Selection ($c_0 < \Delta(e(1))$) becomes harder to satisfy and the equilibrium moves toward the Cost-Deterred Hiding regime. For any fixed hiding cost c_0 , a sufficiently large increase in π drives the equilibrium out of the Partial Pooling region toward full unraveling.*

Proof. See Appendix E. $\square$

Better registry coverage makes silence more informative, because a larger fraction of non-disclosers are hiding by choice rather than by constraint. As a result, buyers discount silence more heavily, concealment becomes less attractive, and the market moves toward disclosure. The testable content is a cross-market comparative static. Markets with higher $\hat{\pi}_m$ should display lower hiding rates $\hat{\sigma}_{0,m}$ and should be disproportionately represented among the markets classified as fully unravelling in Section 4.5.

Proposition 4.5 (Equilibrium effect of c_0) *An increase in c_0 shifts $\Phi(\sigma_0) = \Delta(e(\sigma_0)) - c_0$ uniformly downward. The upper boundary condition for Partial Pooling ($c_0 \leq \Delta(\bar{\theta})$) becomes harder to satisfy and the condition for Cost-Deterred Hiding ($c_0 > \Delta(\bar{\theta})$) becomes easier, so higher hiding costs push the equilibrium toward the Cost-Deterred Hiding regime. In the interior of the Partial Pooling regime, $\partial \sigma_0^*/\partial c_0 < 0$: higher costs reduce the equilibrium hiding rate.*

Proof. See Appendix E. $\square$

The premium in (Eq. 7) generates a transfer from EPC-sensitive buyers to the non-disclosing pool, but it does not accrue symmetrically within that pool. The informed low-quality hider receives the premium but nets zero, because the indifference condition $\Delta(e(\sigma_0^*)) = c_0$ ensures that the gross revenue gain from concealment is exactly consumed by the hiding cost. The uninformed seller, who bears no hiding cost, is the residual claimant and pockets the full premium c_0 as a pure rent. The welfare cost of the equilibrium splits into two distinct components. The first is an allocative distortion. EPC-sensitive buyers pay a positive premium for assets whose true energy performance falls below the pooled belief $e(\sigma_0^*)$, a misallocation that would be eliminated if the concealed information were transmitted. The second is a deadweight loss, since the hiding cost c_0 is incurred by informed sellers at listing time without any counterpart in buyer valuations. This is a social loss only to the extent that c_0 represents real resources destroyed, such as legal fees, prolonged negotiation, or administrative burden. Any portion that is instead a transfer, such as the expected value of a fine accruing to the state, is a redistribution rather than a loss. We do not identify this split, so the deadweight-loss component is an upper bound. A formal efficiency characterization of these two components, and of the optimal subsidy that addresses them, is developed in Section 5.

4.4 Identification

The model leaves two objects to be recovered market by market: ρ_m , the share of buyers who actually price the certificate, and $c_{0,m}$, what it costs an informed seller to keep an unfavourable rating off the listing. Everything else the model needs is something we can count. The ADEME match tells us which sellers hold a certificate (π_m), what those certificates say ($\bar{\theta}_m$), and how often a seller holding a poor one lists without it ($\sigma_{0,m}$). The identification problem is therefore two parameters wide, and the two parameters are reached by different routes.

Three ingredients do the work. The July 2021 reform supplies the price variation: differencing each market's cross-group price gaps around a known date turns them into demand-side objects, purged of the pre-existing level differences that separate one local market from another. The registry supplies the beliefs, because we observe about the non-disclosing pool exactly what a rational buyer would be forming beliefs about, so the equilibrium belief e_m^* can be computed rather than estimated. The third ingredient sits outside the price system altogether. It is the certified energy consumption of the dwellings themselves, and it is what allows buyer sensitivity to be separated from the size of the underlying quality difference. Without it, ρ_m cannot be recovered at all, for a

reason internal to the model rather than a shortcoming of the data: the sensitive-buyer share enters every price the model generates in one place only, so it cancels out of any ratio of price gaps one might form. The concealment cost is the easier of the two. It follows from the disclosed-price gap and the observed hiding rate, and needs neither the anchor nor ρ_m .

4.4.1 The July 2021 Reform as a Natural Experiment

The change in the legal status of the DPE described in Section 3.1 is the source of identification: it created a precise, legally documented change in the value of a disclosed EPC score to buyers, with a known implementation date of July 1, 2021. Crucially, the reform operated on the *demand* side, that is, on the weight a disclosed rating carries in buyers' valuations and legal recourse, and not on the seller's cost of withholding. The obligation to provide a DPE at sale, and the compliance and liability exposure attached to coming to market without the required diagnostic, were already in force under loi ALUR (Article L271-4 CCH) well before 2021. These are precisely the components of the concealment cost $c_{0,m}$. What July 2021 changed was not whether a seller must disclose, nor the cost she bears for withholding, but how much a *disclosed* rating now matters to the buyer. The reform therefore shifts ρ_m while leaving $c_{0,m}$, governed by the pre-existing disclosure mandate, unchanged across the reform date. This is what licenses reading the DiD change in the premium as the demand response that identifies ρ_m , holding the hiding cost fixed. Whether the supply side of the market responded concurrently is a maintained assumption we cannot directly test with the data at hand; we return to it below.⁴

The identifying assumption is the standard parallel-trends condition: in the absence of the reform, the cross-group price gaps $\beta_{1,m,t}$ (energy-efficient vs. energy-intensive disclosed properties) and $\Delta_{m,t}$ (undisclosed vs. energy-intensive) would have continued on their pre-reform trajectory. The DiD does not require that buyers were entirely inattentive to EPC scores before July 2021; it requires only that their relative pricing of disclosure groups was not already trending. Any post-reform shift in the gaps, relative to that pre-reform baseline, is then attributable to the change in buyer incentives induced by legal opposability.

The pre-trend evidence supports this assumption directly, and more strongly than a single test would. Both $\beta_{1,m,t}$ and $\Delta_{m,t}$ are statistically flat across all markets in the 33 months before the reform (Figure 8). These two gaps measure distinct economic objects: $\beta_{1,m,t}$ captures buyers' willingness to pay for verified high quality, while $\Delta_{m,t}$ reflects the informational wedge created by strategic non-disclosure. They respond to different primitives of the model. A supply-side confound (a shift in π_m , $\bar{\theta}_m$, or seller composition at the reform date) would generically move these two gaps by different amounts and in potentially different directions, since the mechanisms through which supply composition enters each gap differ. The fact that both gaps are jointly flat before July 2021 and both shift upward simultaneously after the reform, at the same markets and on the same calendar date, is difficult to reconcile with a supply-side story. It is exactly what the demand-side interpretation predicts: legal opposability increases the return to EPC quality for sensitive buyers, raising both the price of verified high quality ($\beta_{1,m,t}$) and the information premium extracted by non-disclosers ($\Delta_{m,t}$) at the same moment.

We acknowledge that supply-side stability (the assumption that π_m , $\bar{\theta}_m$, and $c_{0,m}$ did not shift discontinuously at the reform) is maintained rather than directly tested. Our listing data begin in 2021Q3, so pre-reform ADEME match rates are not observable. The timing of audit decisions provides some reassurance: certified DPE audits are commissioned and completed before listing, with a gap of weeks to months, so the audit stock at any listing date reflects decisions made prior to that date. But this is a maintained assumption, and we treat it as such.⁵

4.4.2 Two Price Gaps from the Reform

The price side of the identification rests on a panel of monthly hedonic price indices, constructed in Chapter 1 from the same listing data, that tracks three disclosure groups separately within each local market. The first

⁴An alternative strategy based on a Bartik instrument (interacting wholesale gas price variation with the pre-reform share of low-efficiency properties per market) fails at the first stage (F -statistic: 1.0 to 1.6). The French *bouclier tarifaire* (energy price shield, October 2021 to 2023) decoupled household energy costs from the TTF gas price spike driven by the Russian invasion of Ukraine, removing the demand-side variation the instrument was designed to exploit.

⁵Because the disclosure obligation and its associated liability predate the reform (loi ALUR, Article L271-4 CCH), we treat $c_{0,m}$ as governed by that long-standing mandate and unchanged across July 2021. We do not sign the bias a violation of this assumption would introduce into the anchored estimates below, since the anchor is constructed independently of the reform-induced price changes and the direction would depend on exactly how a supply-side shift propagates into the disclosed-price gap $\hat{\beta}_{1,m}$ — a question we leave open rather than resolve by assumption.

covers disclosed energy-efficient properties (grades A to D), the second disclosed energy-intensive ones (grades E to G), and the third every listing that shows no rating at all, a pooled group that necessarily blends strategic hidiers with genuinely uninformed sellers. Each index is the predicted log price per square metre of a reference property, conditional on hedonic controls, so the distance between two of them measures the price wedge attributable to the certificate rather than to differences in what is being sold.

Two of those distances matter. The energy-efficiency premium $\beta_{1,m,t}$ separates the efficient from the intensive disclosed index; the non-disclosure premium $\Delta_{m,t}$ separates the silent index from the intensive disclosed one. Their reform-induced shifts, $\hat{\beta}_{1,m}$ and $\hat{\Delta}_m$, are the differences between post- and pre-reform means of these two series, with the pre-period running from October 2018 to June 2021 and the post-period from July 2021 to July 2024. A volume filter requiring at least twenty transactions per month per disclosure group retains 81 apartment markets and 114 house markets. Markets whose disclosed-price gap fails to turn positive after the reform ($\hat{\beta}_{1,m} \leq 0$) are set aside before the structural inversion, which requires a positive quality premium to invert, leaving 49 apartment markets and 104 house markets (Table 4).

The same panel supports a month-by-month event study, normalised to zero in the last pre-reform month and plotted in Figure 8 over the 33 months before and 36 months after July 2021. Both gaps are statistically indistinguishable from zero throughout the pre-period, which is the parallel-trends evidence discussed above, and both rise gradually from the reform date onward, consistent with progressive capitalisation of a rating that only became legally opposable at that moment. The period means used in estimation are averages of these relative-month profiles over their respective windows.

4.4.3 Beliefs Read Directly from the Registry

The equilibrium belief e_m^* , the quality a buyer rationally attributes to a listing that shows no rating, is normally the hardest object in a disclosure model to take to data: it is a fixed point, and it has to be estimated jointly with everything else. Here it can be read off. The registry gives us, market by market, precisely the three frequencies the belief function depends on — the share of sellers holding a post-reform certificate ($\hat{\pi}_m$), the share of those certificates falling in the efficient band ($\hat{\theta}_m$), and the share of sellers holding an inefficient certificate who nonetheless list without one ($\hat{\sigma}_{0,m}$). Substituting them into the belief function of Section 4 (Equation (Eq. 5), evaluated at $\sigma_1^* = 0$ by Corollary 4.1) returns $\hat{e}_m^*$ directly:

$$\hat{e}_m^* = \frac{(1 - \hat{\pi}_m) \hat{\theta}_m}{\hat{\pi}_m (1 - \hat{\theta}_m) \hat{\sigma}_{0,m} + (1 - \hat{\pi}_m)} \quad (\text{Eq. 8})$$

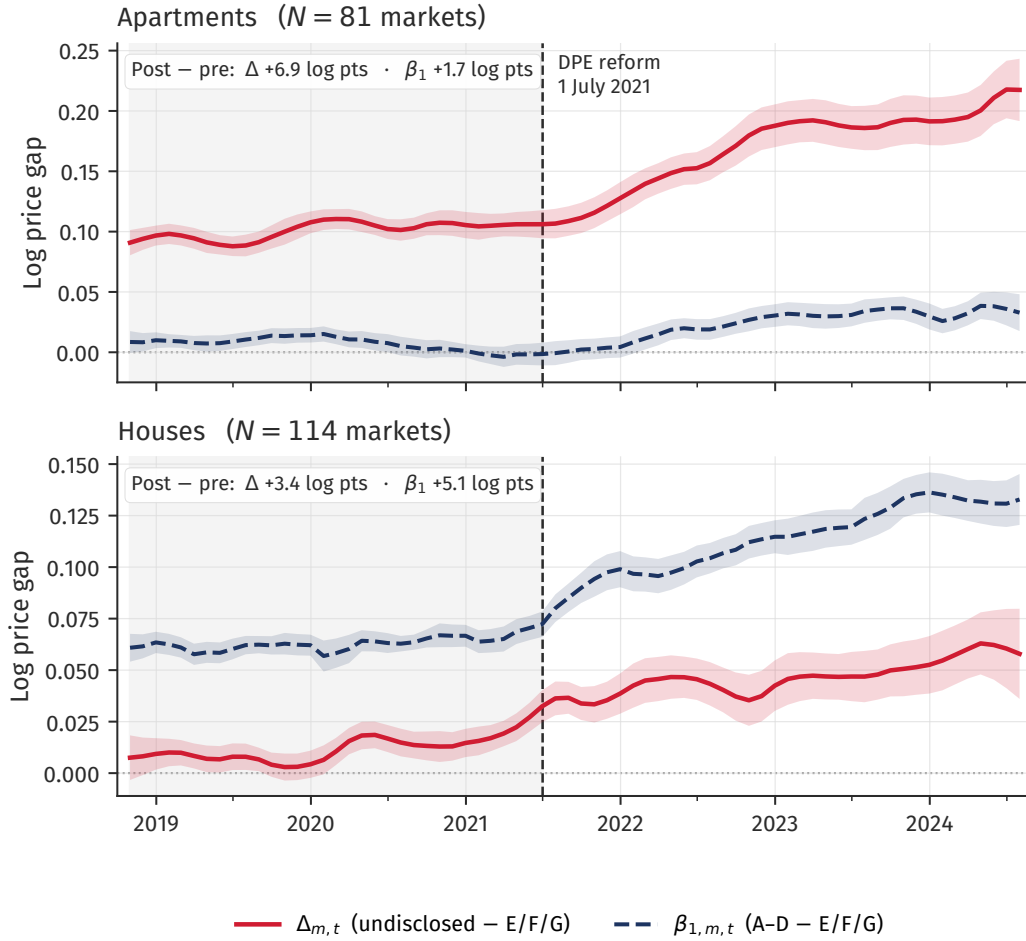
the uninformed high-quality mass sitting in the non-disclosing pool, divided by the total mass of that pool. Nothing is estimated at this step; $\hat{e}_m^*$ is a function of observed sample proportions from the four-group classification of Section 3.2.

What this number tracks is why silence is not damning. The pool of silent listings is dominated by sellers with no certificate to withhold, and their presence holds the quality a buyer infers from silence well above what it would be if every silent seller were concealing something. The registry lets us measure that dilution rather than assume it.

4.4.4 An Energy-Cost Anchor for Buyer Sensitivity

With $\hat{\beta}_{1,m}$, $\hat{\Delta}_m$ and $\hat{e}_m^*$ all in hand, the premium equation $\Delta_m = \rho_m \beta_{1,m} e_m^*$ (Equation (Eq. 7)) appears to deliver buyer sensitivity by rearrangement, as the ratio $\hat{\Delta}_m / (\hat{\beta}_{1,m} \hat{e}_m^*)$, formed either in levels or in reform-induced changes. It does not, and the reason is internal to the model rather than a shortcoming of the data. Buyers who price the certificate and buyers who ignore it are aggregated into a single market price, so the sensitive-buyer share never appears on its own. It appears multiplied by the physical size of the quality difference, and only ever in that combination. The disclosed-price gap equals that product; the gap between silent and intensive listings equals the same product scaled by the pool belief. Their ratio is therefore the pool belief itself, which is exactly what the construction then divides by, so it returns approximately one whatever the true ρ_m happens to be. Pooling markets buys nothing, since every market contributes the same degenerate identity. What the ratio does measure is whether the model's own restriction $\Delta_m = \beta_{1,m} e_m^*$ holds in the data, which makes it a specification test rather than an estimator. Appendix B states and proves this formally, and Figure 9 shows what it looks like in the

Figure 8. Event-study dynamics of the two DiD price gaps around the July 2021 reform



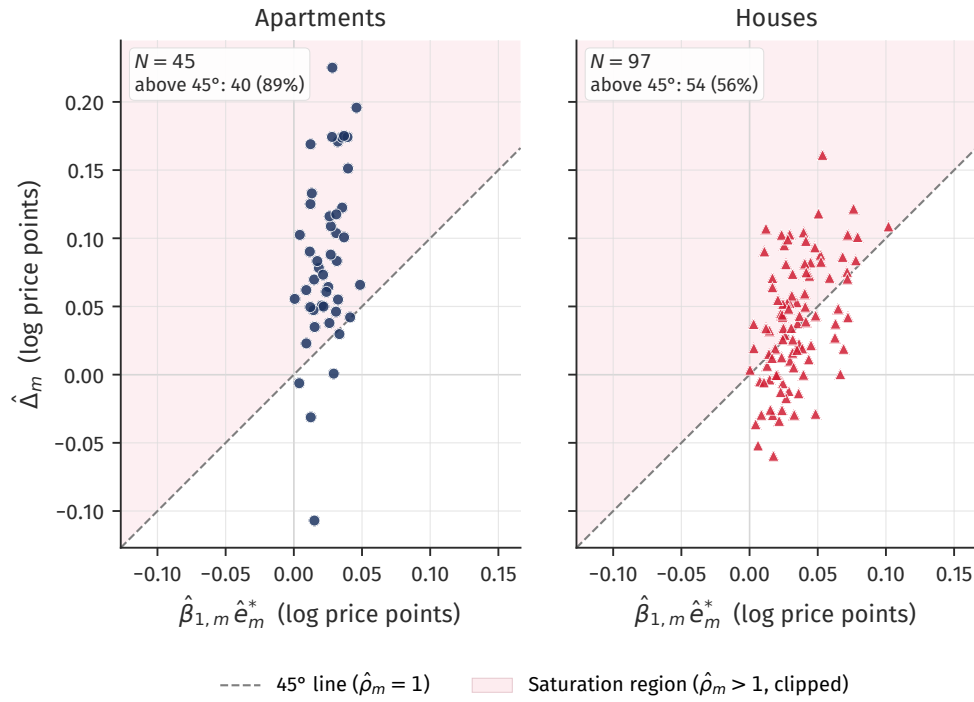
Note. Monthly log price gaps relative to disclosed E/F/G listings, averaged across markets, from the hedonic price-index panel described in Section 4.4. $\Delta_{m,t}$ (solid) contrasts undisclosed listings; $\beta_{1,m,t}$ (dashed) contrasts disclosed A–D listings, both against disclosed E–G. Shaded bands are 95% confidence intervals for the movement of each gap relative to its own pre-reform mean (not the cross-market level, which differences out of the DiD and would otherwise overstate the uncertainty of the trajectory itself). Vertical line: July 1, 2021. The post-minus-pre contrast of each series is the corresponding market-average DiD estimate reported in Table 4, Panel A.

data: apartment markets sit almost entirely above the line where $\hat{\Delta}_m$ would exceed $\hat{\beta}_{1,m}\hat{e}_m^*$, which is exactly the boundary the ratio construction has to clip against in nearly every market, not a handful of outliers.

Separating the two therefore requires information the price system does not contain, and we take it from the physical energy performance of the dwellings themselves. Every ADEME-matched listing carries a certified primary-energy consumption figure in kilowatt-hours per square metre per year, so within each market we can compute the average consumption gap between dwellings certified in the intensive band and those certified in the efficient one. Valued at an energy tariff and discounted over the period a buyer expects to own the property, that gap is the running-cost saving an efficient dwelling delivers. Expressed relative to the local price level, it is what a buyer who attends to the energy bill, and to nothing else the certificate conveys, should be prepared to pay extra for verified efficiency. This benchmark, $\hat{\beta}_{1,m}^{\text{anchor}}$, is built entirely from certified consumption records and an energy price. No listing price enters it, which is what makes it an independent source of identification rather than a rearrangement of moments we have already used.

Converting the consumption gap into euros requires three calibrated inputs: an energy price of about €0.21 per kWh (CRE/Eurostat, 2022–2024 average, blended across the national heating-vector mix), a discount rate

Figure 9. Why the naive ratio saturates: $\hat{\Delta}_m$ against $\hat{\beta}_{1,m}\hat{e}_m^*$, by market



Notes. Each point is a market. The 45° line marks $\hat{\Delta}_m = \hat{\beta}_{1,m}\hat{e}_m^*$, i.e. where the naive ratio $\hat{\Delta}_m/(\hat{\beta}_{1,m}\hat{e}_m^*)$ equals one. Points in the shaded region are markets where that ratio exceeds one and the model's over-identifying restriction (Corollary B.1) fails — not markets with unusually high buyer sensitivity, since ρ_m never enters the plotted quantities. 40 of 45 apartment markets (89%) and 54 of 97 house markets (56%) fall in this region.

of 3.75% (the contemporary French mortgage rate), and a ten-year horizon. The horizon is shorter than full capitalisation of the energy stream would justify, and deliberately so. It matches the typical holding period, and the energy-efficiency-gap literature finds buyers behaving in revealed preference as though they discount durable energy attributes over exactly such truncated horizons (Allcott and Greenstone, 2012). The consumption averages pool disclosed and withheld matched listings, since restricting them to disclosed listings would condition on the seller’s own hiding decision and reintroduce the selection the anchor exists to avoid. Assumption B.1 and Equation (Eq. 15) in Appendix B give the construction in full.

Given the anchor, both structural parameters follow in closed form:

$$\hat{\rho}_m = \frac{\hat{\beta}_{1,m}}{\hat{\beta}_{1,m}^{\text{anchor}}}, \quad \hat{c}_{0,m} = \hat{\beta}_{1,m} \cdot \hat{e}_m^* (\hat{\sigma}_{0,m}). \quad (\text{Eq. 9})$$

Buyer sensitivity is the fraction of the energy-bill benchmark that the market actually pays. The concealment cost is the disclosed-price gap scaled by the belief prevailing at the observed hiding rate, and the second of these expressions uses neither the anchor nor ρ_m . It is the equilibrium indifference condition read backwards: in partial pooling the marginal informed hider is exactly indifferent between disclosing and staying silent, so the revenue she gives up by remaining in the pool must be what silence costs her. One free parameter is matched to one observed moment — the hiding rate among informed low-quality sellers, a group the registry identifies precisely and which is therefore free of the uninformed sellers who contaminate the raw non-disclosure premium — and the match is exact. Appendix B sets out both arguments (Propositions B.2 and B.3).

The anchor also turns an assumption into a test. Since ρ_m is a share, the model requires $\rho_m \leq 1$, which under (Eq. 9) means the observed disclosed-price gap must not exceed the energy-bill benchmark. The data can now refuse this, and mostly do not: the inequality fails in 1 of 45 apartment markets and in 5 of 97 house markets. In the overwhelming majority of markets, the energy-cost saving is large enough on its own terms to rationalise the price advantage certified efficiency commands, without appealing to universal buyer sensitivity. The ratio construction offers no comparable check, because it places every estimable market at the bound by construction.

One prediction of the model plays no part in the estimation and can therefore be tested against it. In partial pooling, the indifference condition implies that the raw non-disclosure premium should equal the concealment cost exactly. Because $\hat{c}_{0,m}$ is built from a different moment, the difference between the two, $\hat{\lambda}_m \equiv \hat{\Delta}_m - \hat{\beta}_{1,m} \hat{e}_m^*$, is a genuine over-identifying restriction (Lemma B.1, Corollary B.1). We report it rather than absorb it into the cost estimate. This is also why we do not simply set $\hat{c}_{0,m} = \hat{\Delta}_m$, which the indifference condition would seem to license: that shortcut imposes the restriction we want to examine. For houses the wedge is small and statistically indistinguishable from zero (mean 0.4 log points, $t = 1.10$), and the restriction survives. For apartments it is large and precisely estimated (mean 6.1 log points, $t = 6.78$), and the restriction is decisively rejected — the premium non-disclosers command in apartment markets substantially exceeds what the disclosed-price gap and the registry-measured belief say it should be. We treat that rejection as a genuine specification tension rather than as noise, and return to it in Section 4.5 and Appendix A.

The structural estimates use listing-level data from 2022 to 2024.⁶ The estimation sample comprises 45 apartment markets and 97 house markets.

4.5 Estimates

4.5.1 Main Estimates

Table 4 reports market-level estimates for both property types. Panel A covers the DiD estimates from the price index panel; Panel B covers the two-step structural estimates; Panel C covers supply-side observables from the ADEME match.

Table 4 yields a coherent and interpretable picture across all three panels. In Panel A, the energy-efficiency premium $\hat{\beta}_{1,m}$ is approximately 1.7% for apartments and 5.1% for houses: disclosing an energy-efficient EPC score commands that price advantage over disclosing an energy-intensive score, after the July 2021 reform. The non-disclosure premium $\hat{\Delta}_m$ (6.9% for apartments and 3.4% for houses) measures how much more non-disclosers

⁶The year 2021 is excluded from the listing-level analysis because the market had not yet fully repriced the reform: the pooled hedonic estimate of $\hat{\beta}_1$ for apartments in H2 2021 is approximately -0.012 , indicating the EPC quality premium had not yet turned positive, which renders the structural equation $\Delta_m = \rho_m \cdot \beta_{1,m} \cdot e_m^*$ unreliable for that period.

Table 4. Market-level estimates by property type

Parameter	Apartments			Houses		
	Mean	Std	N	Mean	Std	N
<i>Panel A: DiD estimates from price index panel</i>						
$\hat{\beta}_{1,m}$ (energy-efficiency premium)	1.7%	3.5	49	5.1%	4.0	104
$\hat{\Delta}_m$ (non-disclosure premium)	6.9%	6.3	49	3.4%	4.8	104
<i>Panel B: Anchored structural estimates, pooled 2022 to 2024</i>						
$\hat{\rho}_m$ (buyer EPC-sensitivity)	49.5%	25.0	45	45.8%	26.7	97
$\hat{c}_{0,m}$ (hiding cost)	2.4%	1.2	45	3.5%	2.0	97
$\hat{\lambda}_m$ (over-ID wedge, $\hat{\Delta}_m - \hat{\beta}_{1,m}\hat{c}_{0,m}^*$)	6.1%	—	45 ($t = 6.78$)	0.4%	—	97 ($t = 1.10$)
<i>Panel C: Supply-side observables from ADEME match</i>						
$\hat{\pi}_m$ (registry-matched share)	50.7%	7.4	45	41.9%	4.4	97
$\hat{\theta}_m$ (mean quality $K=1$)	62.5%	8.8	45	60.5%	10.4	97
$\hat{\sigma}_{0,m}$ (informed hiding rate)	15.2%	9.8	45	6.8%	5.9	97

Notes. Panel A: market-level DiD estimates from the price index panel (pre: Oct 2018 to Jun 2021; post: Jul 2021 to Jul 2024), volume-filtered at ≥ 20 transactions/month/EPC category, $N = 45$ apartment / $N = 97$ house markets with $\hat{\beta}_{1,m} > 0$ that also carry a valid registry-based structural estimate (Panels B–C); a further 4 apartment / 7 house markets pass the DiD filter alone (raising Panel A's own N to 49/104) but lack the ADEME coverage Panels B–C require. $\hat{\Delta}_m$ is no longer treated as an upper bound for $c_{0,m}$; see Panel B's $\hat{\lambda}_m$ row. Panel B: anchored structural estimates (Equation (Eq. 9), Appendix B). $\hat{\rho}_m = \hat{\beta}_{1,m} / \hat{\beta}_{1,m}^{\text{anchor}}$, not clipped to $[0, 1]$ (1 apartment / 5 house markets violate the bound; see Appendix B); $\hat{c}_{0,m} = \hat{\beta}_{1,m} \cdot \hat{e}_m^*(\hat{\sigma}_{0,m})$, independent of ρ_m . $\hat{\lambda}_m$ is the mean over-identifying wedge with its t -statistic against $H_0 : \text{mean}(\hat{\lambda}_m) = 0$ (a simple test, not optimal-GMM); it is not folded into $\hat{c}_{0,m}$. Panel C: supply-side moments computed as sample frequencies from the ADEME match. Means in Panel A and the $\hat{c}_{0,m} / \hat{\lambda}_m$ rows of Panel B are percentage changes (computed as $e^\delta - 1$, expressed in percent). Share parameters in Panels B and C ($\hat{\rho}_m$, $\hat{\pi}_m$, $\hat{\theta}_m$, $\hat{\sigma}_{0,m}$) are in percentage points. Standard-deviation columns are in the same units as the mean (percentage points).

earn over low-quality disclosed properties compared with the pre-reform baseline; we no longer read this as an upper bound on the concealment cost, since $c_{0,m}$ is now identified independently of Δ_m (Panel B; see the $\hat{\lambda}_m$ discussion below). Panel B translates the disclosed-price gap into seller-side parameters via the energy-cost anchor: $\hat{\rho}_m = 49.5\%$ for apartments and 45.8% for houses indicate that roughly half of buyers in the average market price energy performance into their offers — a far more moderate, and far more plausible, reading than the near-universal sensitivity a ratio-based estimator mechanically returns (Appendix B) — while the identified hiding cost $\hat{c}_{0,m}$ settles at 2.4% of property value for apartments and 3.5% for houses. These remain economically material magnitudes once set against the transaction costs a French buyer already faces. Notary fees on an existing dwelling run to roughly 7 to 8% of the sale price, and lenders ordinarily require a personal contribution of at least 10%.⁷ The cost of withholding a certificate is therefore of the order of a third of the notary bill, and roughly a quarter to a third of the minimum deposit a bank would demand: not a rounding error in the transaction, though well short of the sums that routinely change hands around it. That two segments trading very differently land within roughly one percentage point of each other reinforces the reading that this cost is set by the disclosure regime rather than by the property. Panel C turns to the supply side: just over half of apartment sellers (50.7%) and 41.9% of house sellers carry a post-2021 registry record at listing ($\hat{\pi}_m$), and among those, roughly six in ten hold a property rated A–D ($\hat{\theta}_m = 62.5\%$ and 60.5%). The most striking contrast across property types lies in the strategic-hiding rate $\hat{\sigma}_{0,m}$: 15.2% of informed apartment sellers who hold an E/F/G rating choose to withhold it, versus 6.8% for houses, a difference that mirrors the three-fold gap in $\hat{\beta}_{1,m}$, since a larger forgone quality premium

⁷Transfer duties (*droits de mutation à titre onéreux*) were raised to a maximum of 6.31% for transactions concluded from April 2025, an increase adopted by 84 of the 101 *départements*, with first-time buyers exempt; total notary fees on an existing dwelling accordingly sit near 7 to 8% of the price. The 10% personal contribution is not a legal requirement but the threshold applied by most lenders, and is intended to cover precisely these ancillary costs.

makes concealment a costlier choice for house sellers.

4.5.2 *Structural Estimates and Economic Interpretation*

Buyer sensitivity emerges from the anchor as a genuine point estimate in each market rather than as a value pressed against a boundary. About half the buyers in the average market price energy performance into their offers — 0.495 for apartments, 0.458 for houses — and markets spread around those averages in a way the estimator does not manufacture, with a cross-market standard deviation of 0.25 and 0.27 respectively. A ratio-based construction returns nothing of the kind; it piles the whole distribution onto the upper bound. That bound is now something the data can refuse. Since the sensitive share cannot exceed one, the anchored estimate requires the disclosed-price gap to stay below the energy-bill benchmark, and the requirement fails in only 1 of 45 apartment markets and 5 of 97 house markets. Almost everywhere, the running-cost saving an efficient dwelling delivers is large enough on its own terms to account for the price advantage certified efficiency commands, without supposing that every buyer is paying attention.

One consequence of the closed-form estimator is worth stating plainly, because it closes off a finding the model might otherwise appear to offer. The concealment cost is the disclosed-price gap scaled by a belief that is itself confined, by construction, between the value it would take if every informed low-quality seller hid and the value it would take if none did. The cost can therefore never land outside the Partial Pooling band, and every market in the sample is classified as Partial Pooling. Cost-Deterred Hiding and Complete Adverse Selection remain possible corner regimes in the model, but a market can only be sorted into one of them when buyer sensitivity has been clipped to a boundary. With sensitivity point-identified, that sorting cannot arise here, and we report no regime heterogeneity as an empirical finding.

The cost of keeping an unfavourable rating off the listing is economically material and strikingly similar across the two segments: 0.024 log points, roughly 2.4% of property value, for apartments, and 0.035 log points, or 3.5%, for houses. Two segments that otherwise trade very differently landing within about a percentage point of each other points to a cost set by the legal disclosure regime rather than by the property.

Nothing in this construction forces the concealment cost to sit below the non-disclosure premium, and it need not. The two are built from different moments, and the distance between them is the model's over-identifying restriction (Lemma B.1), which we report rather than resolve by fiat. In house markets that wedge averages 0.4 log points and cannot be distinguished from zero ($t = 1.10$), so the restriction survives. In apartment markets it is 6.1 log points and precisely estimated ($t = 6.78$): non-disclosers there command substantially more than the disclosed-price gap and the registry-measured belief say they should under Partial Pooling. Truncating the post-reform window to one or two years instead of the full three leaves both results standing (Appendix A), so neither turns on the endpoint we happened to choose. If anything the two move apart as the window lengthens, the apartment wedge widening and the house wedge narrowing further, a divergence rather than a coincidence of the final year of data.

The apartment wedge deserves a closer look, because more than one channel could produce it without disturbing the estimates reported above. One candidate is measurement rather than economics. Apartments are the only property type in this sample where a certificate can in principle rest on a building-level diagnostic rather than a unit-specific one, and matching a listing to the registry is inherently harder within a single multi-unit address than against a house's unique street address. Ordinary noise of that kind in the certified-quality signal attenuates the disclosed-price gap $\hat{\beta}_{1,m}$ toward zero, and since the concealment cost is built as $\hat{\beta}_{1,m} \cdot \hat{e}_m^*$, a smaller $\hat{\beta}_{1,m}$ mechanically pulls $\hat{c}_{0,m}$ down and widens the wedge, which is the direction the apartment numbers actually move in. A second candidate is coarseness rather than noise. A single administrative market plausibly spans a wider range of building vintage, heating system, and floor level among apartments than among houses drawn from the same geography, and buyers can read part of that variation directly, without a certificate. A belief built only from registry frequencies then misses what those substitute signals already tell apartment buyers, and the model attributes the resulting gap to the concealment cost rather than to information the certificate was never the only source of. Both stories point at the data the estimator is fed rather than at the identification argument itself, and neither moves $\hat{\rho}_m$ or $\hat{c}_{0,m}$ as reported.

Two further candidates are economic rather than mechanical, and they are the more interesting ones because they would mean the model is too parsimonious in this segment rather than wrong. Apartment markets carry a

higher share of informed sellers and a higher hiding rate than house markets, both consistent with a larger investor or professional-seller presence in dense urban stock. If buyers price not only the average concealment cost but also their uncertainty about who is behind a given silent listing, an uninformed owner-occupier or a professional concealer, a silent apartment carries a risk premium that the single mean-cost parameter c_0 has no room for. The other candidate follows from how apartment markets actually clear. Buyers in the thick, fast-moving urban markets where most of our apartment sample sits have less room to wait out a non-disclosing seller than buyers in the more dispersed markets houses trade in, and a seller who can extract more from a thinner buyer bargaining position earns a premium the model's belief-driven channel does not carry. We do not resolve which of these forces, or what combination, accounts for the apartment wedge here, and we do not need to. The point-identified $\hat{\rho}_m$ and $\hat{c}_{0,m}$ this section reports stand on the anchor and the registry match alone, and the wedge is a diagnostic on top of them rather than a correction to them. We return to the market-thinness channel in future work, where it is the object of study rather than a residual.

The contrast between the property types is itself informative. The energy-efficiency premium in house markets runs roughly three times what it does in apartments, 0.051 against 0.017 log points. We do not test why, and more than one channel fits. Houses are larger and older on average, so heating weighs more heavily in a buyer's running expenses. They also sit predominantly outside city centres, in markets where demand pressure is weaker and price formation depends less on locational scarcity, which leaves more of the price free to respond to an attribute such as energy performance. In tight urban apartment markets the opposite holds: location dominates valuation and compresses the share any single characteristic can claim. Separating these explanations would require variation we do not exploit here.

What follows from the larger premium is easier to pin down. A seller who hides a poor rating in a house market forgoes more by doing so, and the data reflect that exactly. The hiding rate among informed low-quality sellers is well under half the apartment figure, 6.8% against 15.2%, and strategic hidiers make up 1.1% of house listings against 3.5% of apartment listings. The non-disclosure premium follows in turn, 0.034 for houses against 0.069 for apartments, because a thinner pool of strategic hidiers leaves buyers less pessimistic about what silence conceals. Higher quality premium, lower hiding rate, lower non-disclosure premium: the whole pattern is what the model's equilibrium comparative statics predict when markets differ in the return to verified quality.

5 Policy Extension: Renovation Subsidies

5.1 Motivation

The core model treats θ as exogenous, but the policy question is what happens once it becomes a choice. An informed seller who holds a low-quality certificate has three ways to respond to the disclosure regime: pay c_0 to hide it and hope to share in the non-disclosure premium, disclose the low score and take the discount that comes with it, or renovate at engineering cost $R(X)$ and disclose a verified high score instead. What renovation has to clear is the verified-quality premium, the price gap between a disclosed high-quality listing and a disclosed low-quality one. Whenever renovating costs more than that premium, the seller declines to invest, and a subsidy is the natural instrument for closing the gap.

A subtler point runs through the rest of this section, and it is worth flagging before the formal argument: the renovation threshold looks, at first pass, as though it should rise one-for-one with the hiding cost c_0 , and it does not. The reason starts with which alternative renovation is actually competing against. An informed low-quality seller who is weighing renovation is not really choosing between renovating and disclosing her true low quality; she is choosing between renovating and hiding, because disclosing low quality is itself a strategy no informed seller would take over hiding once the non-disclosure pool carries a positive premium. Hiding delivers the inflated pool price at cost c_0 , and that premium net of cost is weakly positive whenever anyone hides in equilibrium at all, so disclosing low quality is weakly dominated and drops out of the comparison. The threshold that remains is pinned by renovate-versus-hide, and on its face the hiding cost enters that threshold additively, as something a seller escapes by renovating. But the hide payoff is not a primitive: it is an equilibrium object, and it moves with c_0 through the equilibrium hiding rate that c_0 itself helps determine. In the partial-pooling regime, the same indifference condition that pins that hiding rate forces the hiding-cost terms to cancel exactly, and the threshold collapses to the verified-quality premium alone, with no residual dependence on c_0 (Lemma 5.1, proven in Appendix E). The regulator nevertheless retains a meaningful disclosure-policy lever: *verification production*, the rate at which sellers obtain EPC audits and enter the informed population π . This margin does not move

the marginal seller's renovation threshold either, but it scales the population of sellers subject to that threshold and, over time, converts the housing stock from uninformed to verified status. Renovation subsidies, hiding-cost enforcement, and verification enforcement therefore act on three different margins of the seller's problem, and are best understood as complements rather than substitutes.

5.2 Extended Seller Problem

Formalizing this requires one addition to the model: a renovation choice $r \in \{0, 1\}$ that upgrades an informed low-quality seller's true type to $\theta' = 1$ when $r = 1$. Her extended problem is

$$\max_{\{r, d, p^{\text{list}}\}} \Pi(r, d, p^{\text{list}} \mid \theta, X) = \phi(p^{\text{list}}) \cdot \Gamma(p^{\text{list}} \mid d, \theta') - C(\theta', d) - (R(X, r) - S) \quad (\text{Eq. 10})$$

where a subsidy $S \geq 0$ reduces the effective renovation cost she faces.

Three strategies survive scrutiny, laid out in Table 5. A seller can disclose the low score as is, hide it, or renovate and disclose the resulting high score. A fourth combination, renovating and then hiding, is never worth considering, and Corollary 4.1 (proven in Appendix E) is exactly the result that rules it out. A verified $\theta = 1$ seller strictly prefers to disclose, whatever the prevailing pool belief, so a seller who has already paid to renovate has no reason to then conceal what she paid for.

Table 5. Strategies for an Informed Low-Quality Seller ($\theta = 0$)

Strategy	Renovate	Disclose	Buyer observes	Gross revenue	Extra cost
1. Disclose $\theta = 0$	No	Yes	$\theta = 0$ verified	$\Pi(d = 1, \theta = 0)$	0
2. Hide	No	No	$e(\sigma_0^*)$ (pool quality)	$\Pi(d = 0)$	c_0
3. Renovate & disclose	Yes	Yes	$\theta = 1$ verified	$\Pi(d = 1, \theta = 1)$	$R(X) - S$

Notes: "Extra cost" is measured relative to Strategy 1 (baseline disclosure at $\theta = 0$). The gross revenue terms $\Pi(d, \theta)$ absorb the transaction cost $C(\theta', d)$ from eq. (Eq. 10) and are distinct objects. $e(\sigma_0^*)$ denotes the equilibrium pool quality belief (Eq. 5). The strategy ($r = 1, d = 0$) is omitted as strictly dominated (Corollary 4.1).

Table 5 also makes plain why Strategy 1 does not survive as a live alternative once Strategy 2 is on the table. Whenever the non-disclosure pool carries a positive opacity premium $\Delta(e^*) = \rho\beta_1 e^* > 0$, an informed low-quality seller prefers hiding to disclosing her true quality outright, since the revenue gain from sitting in the non-disclosure pool at least covers what hiding costs her. In any equilibrium where some sellers hide at all, the equilibrium hiding rate satisfies $\Pi(d = 0; e^*) - c_0 \geq \Pi(d = 1, \theta = 0)$, with equality exactly at the partial-pooling indifference point. Disclosing low quality is therefore weakly dominated by hiding, and the renovation decision collapses to a single comparison between Strategy 3 (renovate and disclose) and Strategy 2 (hide). (Renovating also dominates Strategy 1 directly whenever $R(X) - S < \Pi(d = 1, \theta = 1) - \Pi(d = 1, \theta = 0)$, but the policy-relevant margin is the transition away from Strategy 2, since the concealment trap binds only for sellers who are already hiding.) Renovating wins whenever the renovation cost net of subsidy stays below a threshold equal to the verified-quality premium plus the hiding cost the seller no longer has to pay. This threshold is the object $\bar{R}(\pi, \rho, X)$, defined and analyzed formally below (eq. (Eq. 11)). Absent a subsidy, renovation is privately optimal whenever $R(X) \leq \bar{R}$; the minimum incentive-compatible subsidy is the shortfall $R(X) - \bar{R}$, the transfer needed to restore private incentives once the engineering cost outruns what the market rewards.

What makes $\bar{R}$ interesting for policy is exactly the point flagged in the motivation above. It looks, from the raw definition, as though it should move one-for-one with the hiding cost c_0 , since c_0 enters as something the seller avoids by renovating. That appearance does not survive contact with equilibrium. The pool revenue a hider collects is not fixed; it is evaluated at the equilibrium pool quality $e^* = e(\sigma_0^*)$, and the hiding rate σ_0^* itself adjusts to c_0 . Once that adjustment is taken into account, $\bar{R}$ in the empirically dominant partial-pooling regime turns out to be invariant to c_0 altogether, because the apparent dependence is exactly absorbed by the indifference condition that pins the hiding rate in the first place. The next section makes this precise.

5.3 The Renovation Threshold and the Minimum Incentive-Compatible Subsidy

Definition 5.1 (Renovation Threshold) *The renovation threshold $\bar{R}(\pi, \rho, X)$ is the maximum renovation cost at which an informed low-quality seller prefers to renovate-and-disclose over hiding, absent subsidies:*

$$\bar{R}(\pi, \rho, X) \equiv \underbrace{\Pi(d=1, \theta=1) - \Pi(d=0; e^*)}_{\text{Market premium for verified high quality}} + \underbrace{c_0}_{\text{Hiding cost avoided}}. \quad (\text{Eq. 11})$$

Lemma 5.1 (Equilibrium reduction in partial pooling) *In the partial-pooling equilibrium, the indifference condition $\Pi(d=0; e^*) - c_0 = \Pi(d=1, \theta=0)$ implies that the renovation threshold reduces to the verified-quality premium:*

$$\bar{R}^{PP}(\rho, X) = \Pi(d=1, \theta=1) - \Pi(d=1, \theta=0), \quad (\text{Eq. 12})$$

which depends on ρ and X alone and is invariant to c_0 and π . In Complete Adverse Selection ($\sigma_0^* = 1$), the threshold retains the form (Eq. 11) with $e^* = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$ locally constant in c_0 , so $\partial \bar{R}^{CAS}/\partial c_0 = 1$.

Proof. See Appendix E. $\square$

The partial-pooling reduction (Eq. 12) is the central technical result of this section. It rests on the equilibrium indifference identity: the marginal informed-low seller is, by construction of the regime, exactly indifferent between hiding and disclosing-low, so the hide payoff $\Pi(d=0; e^*) - c_0$ equals the disclose-low payoff $\Pi(d=1, \theta=0)$. Substituting this identity into (Eq. 11) causes the hiding cost to cancel exactly, leaving the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$: the revenue gain a seller obtains by upgrading from a verified-low to a verified-high listing.

In the empirically dominant interior regime, the renovation incentive is governed entirely by the price gap between verified high and verified low quality. The non-disclosure premium $\Delta = \rho\beta_1 e^*$, which in partial pooling equals c_0 by the same indifference, plays no role in the renovation calculus: the seller's gross premium Δ for being in the non-disclosure pool and her cost c_0 for hiding cancel because the regime is constructed to make them cancel. Raising the hiding cost shifts the equilibrium hiding rate σ_0^* along the indifference curve (with $\Pi(d=0; e^*)$ adjusting endogenously to preserve indifference) but does not change the renovation threshold. Policymakers cannot “buy” renovation by tightening hiding-cost enforcement in this regime; renovation requires a direct subsidy that improves the renovate-and-disclose payoff relative to the seller's continuation value at low quality.

Definition 5.2 (Minimum incentive-compatible subsidy) *The minimum incentive-compatible subsidy is the smallest transfer making renovation weakly preferred to hiding, for a seller with renovation cost $\hat{R}(X)$, holding equilibrium beliefs fixed:*

$$S^*(\pi, \rho, X) \equiv \max\{\hat{R}(X) - \bar{R}(\pi, \rho, X), 0\} \quad (\text{Eq. 13})$$

where $\hat{R}(X)$ denotes the actual renovation cost from engineering estimates or program data.

Remark (S^* is incentive-compatibility, not welfare optimum). The object S^* is the minimum transfer that flips the marginal informed-low seller's decision rule from concealment to renovation, given everyone else's equilibrium behaviour. It is *not* a Ramsey-optimal subsidy in the sense of Ramsey (1927). A full welfare problem would maximise

$$\max_S \int_X \mathbf{1}\{\hat{R}(X) \leq \bar{R} + S\} \cdot [\Delta W(X) - \lambda S] dF(X),$$

where $\Delta W(X)$ is the social gain per renovation (private surplus change plus any energy-use externality), $\lambda \geq 1$ is the marginal cost of public funds, and F is the within-market distribution of renovation costs. The corner solution $S^* = \hat{R} - \bar{R}$ emerges only in the limit of degenerate F , $\lambda = 1$, and zero externality. Throughout this section we treat S^* as the operative *positive* object (the smallest subsidy that induces a given seller to renovate) and leave the full Ramsey characterisation to future work.

The renovation threshold $\bar{R}$ is the market's implicit valuation of quality information from the seller's perspective. In the partial-pooling equilibrium it equals the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$, the revenue gain from a buyer-verifiable upgrade from low to high quality. In Complete Adverse Selection it retains the original definition (Eq. 11) with the equilibrium pool quality at the corner. When actual renovation

costs fall below the threshold, the market already rewards quality revelation enough to make private renovation worthwhile. When $\hat{R}(X) > \bar{R}$, the market premium for renovation is insufficient and targeted public support is required. The minimum incentive-compatible subsidy S^* is precisely the difference: it fills the gap between what the market offers and what renovation costs, doing no more than necessary to tip the seller's calculus toward quality improvement.

5.4 Comparative Statics: Renovation and Disclosure Operate on Different Margins

Proposition 5.1 (Subsidy comparative statics by regime) *The minimum incentive-compatible subsidy $S^*(\pi, \rho, X)$ satisfies $\partial S^*/\partial \hat{R}(X) = 1\{\hat{R}(X) > \bar{R}\}$ throughout, together with the following regime-conditional comparative statics on the renovation threshold $\bar{R}$:*

- (i) **Partial Pooling** (the empirically relevant interior regime): $\partial \bar{R}/\partial c_0 = 0$, $\partial \bar{R}/\partial \pi = 0$, and $\partial \bar{R}/\partial \rho > 0$. Hence

$$\frac{\partial S^*}{\partial c_0} = 0, \quad \frac{\partial S^*}{\partial \pi} = 0, \quad \frac{\partial S^*}{\partial \rho} \leq 0 \quad \text{when } \hat{R}(X) > \bar{R}.$$

- (ii) **Complete Adverse Selection** ($\sigma_0^* = 1$, corner): $\partial \bar{R}/\partial c_0 = 1$, $\partial \bar{R}/\partial \pi > 0$, $\partial \bar{R}/\partial \rho > 0$. Hence $\partial S^*/\partial c_0 = -1$ and $\partial S^*/\partial \pi \leq 0$ when $\hat{R} > \bar{R}$.

Proof. See Appendix E. $\square$

The threshold result of Proposition 5.1 should not be read as evidence that disclosure policy is irrelevant for renovation outcomes. It is irrelevant only for one specific decision: the marginal informed low-quality seller's choice between concealment and renovation, holding her informed status fixed. The broader policy problem involves three distinguishable instruments acting on three distinguishable margins.⁸

The first is the intensive margin of renovation propensity. The subsidy S acts directly on the seller's renovation calculus through the inequality $\hat{R}(X) - S \leq \bar{R}$. Conditional on being informed and low quality, the probability of renovation is governed entirely by S and by the cross-property distribution of $\hat{R}(X)$.

The second is the composition of disclosure decisions among the informed. The hiding cost c_0 , raised through fines, audits of suspected concealers, or strengthened legal exposure under loi ALUR, acts on the share of informed-low sellers who hide versus disclose-low. By Proposition 4.3, tightening hiding penalties shrinks the strategic-hiding share. By Lemma 5.1, however, this leaves the renovation threshold unchanged in partial pooling: the equilibrium pool revenue $\Pi(d = 0; e^*)$ adjusts to keep the hider exactly indifferent with the disclose-low alternative, so the c_0 in (Eq. 11) is exactly offset by the equilibrium movement of $\Pi(d = 0; e^*)$. Only in the Complete Adverse Selection corner, where $\sigma_0^* = 1$ is at the boundary and $\Pi(d = 0; e^*)$ is locally constant in c_0 , does c_0 enter the threshold and the textbook substitutability $\partial S^*/\partial c_0 = -1$ obtain.

The third is the extensive margin of verification production. Mandatory EPC enforcement, audit infrastructure investment, and inspection of cadastral records raise π , the share of sellers who are informed about their property's energy performance, which is the principal long-run instrument in France: every transaction since 2007 in principle triggers an EPC, and the ADEME registry grows monotonically over time. The effect on aggregate renovation is direct and first-order:

$$\text{Renovation flow} = \underbrace{\pi}_{\text{informed share}} \cdot \underbrace{(1 - \bar{\theta})}_{\text{low among informed}} \cdot \underbrace{F_R(\bar{R}(\rho, X) + S)}_{\text{share with } \hat{R} \leq \bar{R} + S}, \quad (\text{Eq. 14})$$

where F_R is the within-market distribution of engineering renovation costs. Raising π scales the renovation base linearly even when the per-seller threshold is invariant, because additional sellers become informed and enter the

⁸A natural fourth candidate is ρ itself. The July 2021 reform is direct evidence that policy can shift it (Section 4.4), and the rental-ban extension under *loi Climat et Résilience* is a second channel already in force. We exclude ρ from the instrument set below for a reason of scope, not of principle. Both channels are disclosure-regime and rental-market reforms whose costs and consequences run well outside the renovation-subsidy problem addressed here. Raising ρ is also not confined to the renovation margin. It raises the non-disclosure premium in both corner regimes and widens the Partial Pooling interval itself (Corollary 4.2), effects this policy extension does not price. The three instruments below are the ones the renovation-subsidy problem can target on its own terms.

strategic disclosure problem in the first place. Each newly verified low-quality property is a potential subsidy recipient that previously was not.

There is also a stock-evolution effect that operates only along the verification-production margin. As π rises and audits accumulate, the housing stock becomes increasingly characterised by verified energy labels. Buyers can condition prices on observed θ for a growing share of the market; the uninformed pool that sustains the non-disclosure premium shrinks; and the equilibrium gradually drifts from partial pooling toward cost-deterred hiding as the gain from concealing low quality erodes. Renovations undertaken today, once disclosed at the next sale, become permanent reclassifications from $\theta = 0$ to $\theta = 1$ in the verified stock. The static threshold analysis above takes the share of informed sellers as given; the verification enforcement margin is precisely what makes this share endogenous to policy over the medium run.

The minimum incentive-compatible subsidy $S^*(\pi, \rho, X)$ measures how much transfer it takes to flip the renovation decision of a single informed low-quality seller, given the equilibrium pool composition, and the intensive margin alone determines that object. Hiding-cost enforcement does not substitute for the subsidy in the empirically dominant partial-pooling regime: at the indifference margin, the apparent $+c_0$ in the threshold is exactly cancelled by the equilibrium movement of $\Pi(d = 0; e^*)$, and the renovation incentive collapses to the verified-quality premium. Verification production is genuinely complementary: π does not change the threshold either, but it directly expands the population subject to it and dynamically converts the housing stock toward verified status. Disclosure enforcement and renovation subsidies are therefore complements that scale each other's reach, not substitutes.

Finally, the model suggests that a uniform national subsidy is generically inefficient. The threshold $\bar{R}(\pi, \rho, X)$ varies across markets through ρ (the share of EPC-sensitive buyers) and through property characteristics X , even in partial pooling where π and c_0 drop out of the marginal threshold. A uniform national subsidy therefore necessarily over-subsidises markets with high ρ (where the verified-quality premium is already large) and under-subsidises markets with low ρ (where it is small). The verification-production margin reinforces this: at any given S , the renovation flow (Eq. 14) also scales with the local π , so subsidies and audit-infrastructure investment should be co-targeted on markets where both π is low and ρ delivers a sufficient verified-quality premium to make subsidised renovation cost-effective. Efficient policy requires targeting on observable proxies for market composition: local climate and energy-cost exposure for ρ , audit infrastructure (notaire density, ADEME registry coverage) for π , and property-level engineering estimates for $\bar{R}(X)$.

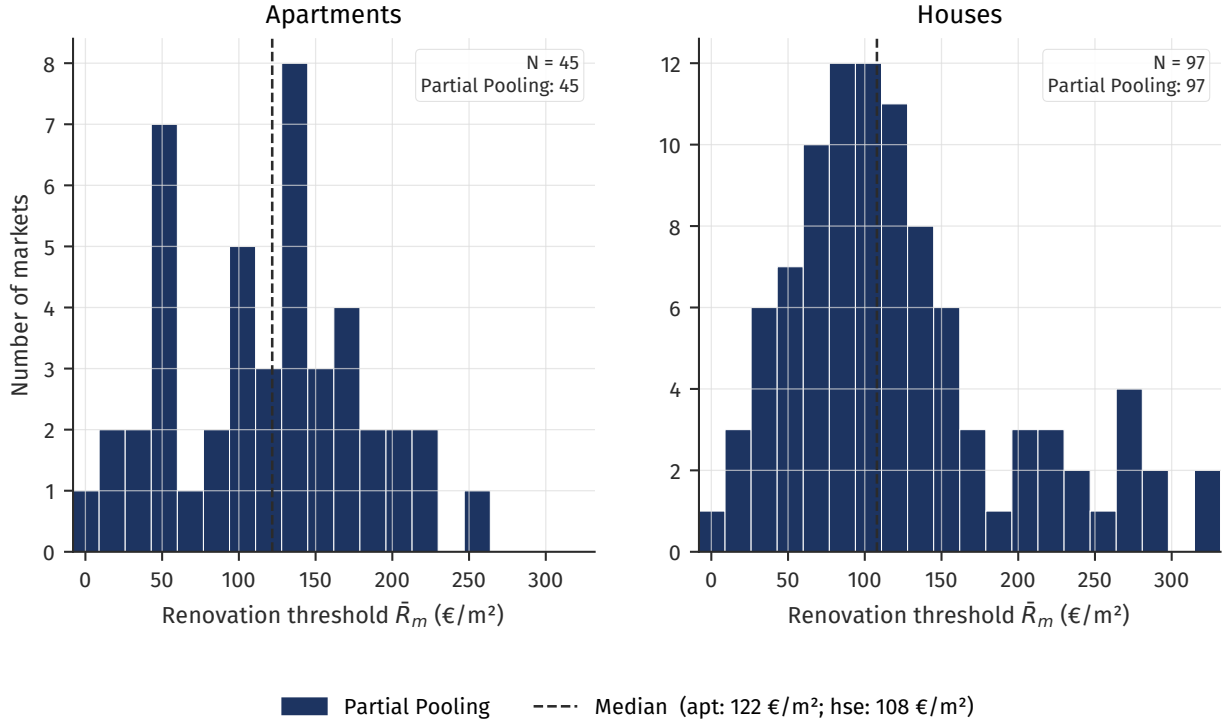
5.5 Renovation Subsidies in French Housing Markets: Empirical Magnitudes and Spatial Heterogeneity

In the partial-pooling regime, Lemma 5.1 reduces the renovation threshold $\bar{R}^{PP}$ to the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$, with c_0 and π cancelling through equilibrium indifference. Because β_1 measures the quality premium in log-point terms against a local price-per-m² base, the nominal threshold $\bar{R}_m$ scales with the local price level. High-price urban markets have the largest verified-quality premia in absolute terms, even when the proportional premium β_1 is similar across locations. Across the 45 estimable apartment markets, $\bar{R}_m$ averages €118/m² (median €122/m²); across the 97 house markets, the mean is €124/m² and the median €108/m² (Figure 10). Every market in our estimation sample sits in the Partial Pooling regime under the anchored estimator, so the threshold is pinned at the verified-quality premium throughout; there is no corner-regime subset with a mechanically larger threshold to separate out. Cross-market dispersion nonetheless remains substantial. Apartment thresholds range from roughly €4 to €253/m², so markets at the upper tail of the distribution are intrinsically more amenable to privately funded renovation than those at the lower tail.

The minimum incentive-compatible subsidy is $S_m^* = \max\{\hat{R} - \bar{R}_m, 0\}$, evaluated here at two renovation-cost benchmarks: €250/m² (a light, “geste par geste” intervention such as roof or wall insulation) and €500/m² (a medium, two-measure upgrade), drawing on MaPrimeRénov’ programme documentation.⁹ At $\hat{R} = 250$, 44 of 45 apartment markets (97.8%) require a positive subsidy; the median S_m^* conditional on $S_m^* > 0$ is €130/m², or approximately €8,186 per unit at the median apartment area of 63 m². For houses, 88 of 97 markets (90.7%) require support, with a conditional median of €147/m² (approximately €16,459 per unit at 114 m²). By $\hat{R} = 500$,

⁹MaPrimeRénov’ distinguishes “geste par geste” interventions (insulation of walls or roof, costing roughly €100–400/m² of treated area) from comprehensive “rénovation d’ampleur” operations (€700–1,200/m² or more); the €500/m² benchmark corresponds broadly to a two-measure upgrade.

Figure 10. Distribution of the renovation threshold $\bar{R}_m$ across markets



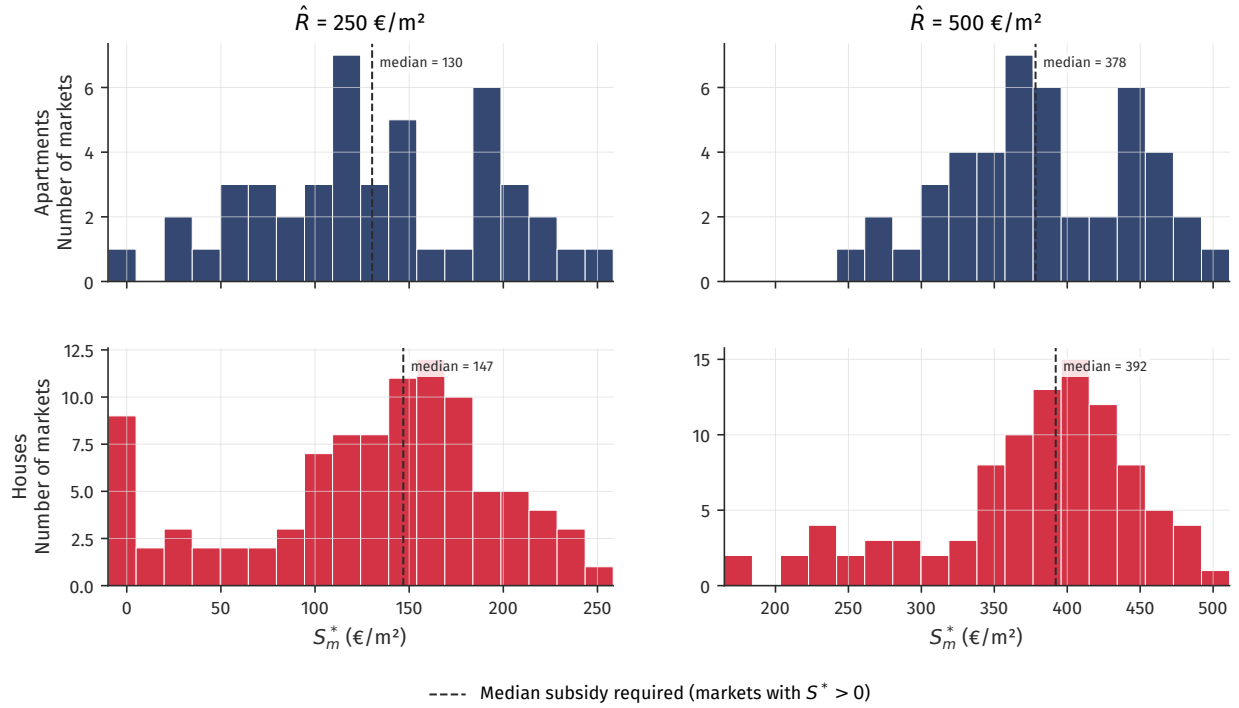
Notes. Each panel shows the cross-market distribution of $\bar{R}_m$ (euros per m²) for one property type. Dashed vertical lines mark within-panel medians. Every market in the estimation sample is in the Partial Pooling regime, so the threshold equals the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$ throughout (Lemma 5.1).

every market of both property types requires a subsidy; the conditional medians rise to €378/m² (apartments, €23,420 per unit) and €392/m² (houses, €43,933 per unit). The cost-sharing ratio $S_m^*/\hat{R}$ is the natural decision variable for the regulator. Among markets requiring support, the mean stands at 54.0% for apartments and 57.0% for houses at the light benchmark; at the medium benchmark both rise to roughly 75–76%. This non-linear escalation in required cost-sharing as intervention depth increases has a direct programme-design implication. A commitment to a fixed rate is more defensible than a commitment to a fixed euro amount, because it adjusts automatically to the depth of the intervention and to local market conditions.

Because every market is in the Partial Pooling regime, the threshold is invariant to c_0 throughout (Lemma 5.1), and cross-market dispersion in S_m^* is driven entirely by variation in the verified-quality premium $\beta_{1,m}$ and local prices, not by any regime-composition effect, since there is no corner-regime subset in our estimation sample to drive one. The policy challenge is correspondingly uniform in structure across markets. The renovation threshold is pinned at the verified-quality premium everywhere, and public intervention must bridge the gap between that premium and the renovation cost market by market.

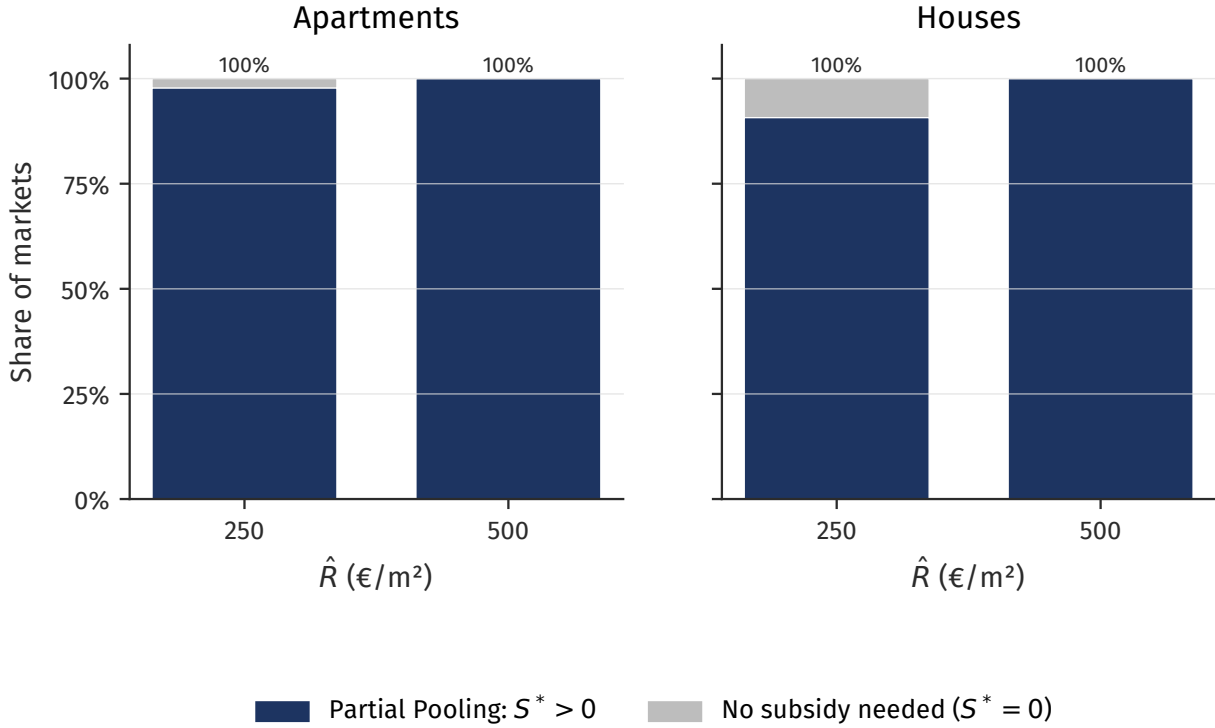
A uniform national subsidy is inefficient because the cost-sharing requirement varies substantially across markets. Figure 13 maps $S_m^*/\hat{R}$, the share of the renovation invoice the regulator must cover, at the two cost benchmarks; bubble size encodes the local verified-quality premium $\bar{R}_m$. The support of the cost-sharing distribution is wide. At $\hat{R} = 250$ €/m², only 2.2% of apartment markets and 9.3% of house markets need no subsidy at all, while markets at the upper end of the distribution require the public purse to cover the entire renovation cost. Among markets that do require support, the mean cost-share is 54.0% for apartments and 57.0% for houses. A flat per-m² grant set at the cross-market mean would simultaneously over-pay the lower tail of this distribution, where the private incentive already suffices, and under-pay the upper tail, where renovations remain unfunded. Both forms of misallocation are costly, the former through fiscal waste on infra-marginal markets and the latter through forgone renovations in the markets where the gap is largest. This cross-market dispersion is systematic rather

Figure 11. Distribution of the minimum incentive-compatible subsidy S_m^* across markets



Notes. Rows correspond to property type (apartments, houses); columns to the renovation-cost benchmark $\hat{R} \in \{250, 500\} \text{ €/m}^2$. Each cell shows the cross-market density of S_m^* (euros per m²). The dashed vertical line within each cell marks the median conditional on $S_m^* > 0$. Markets with $S_m^* = 0$, for which renovation is privately optimal without intervention, appear in the left spike. The benchmark $\hat{R} = 250 \text{ €/m}^2$ corresponds to a light, “geste par geste” intervention; $\hat{R} = 500 \text{ €/m}^2$ to a medium, two-measure upgrade.

Figure 12. Share of markets needing a subsidy, by renovation-cost benchmark

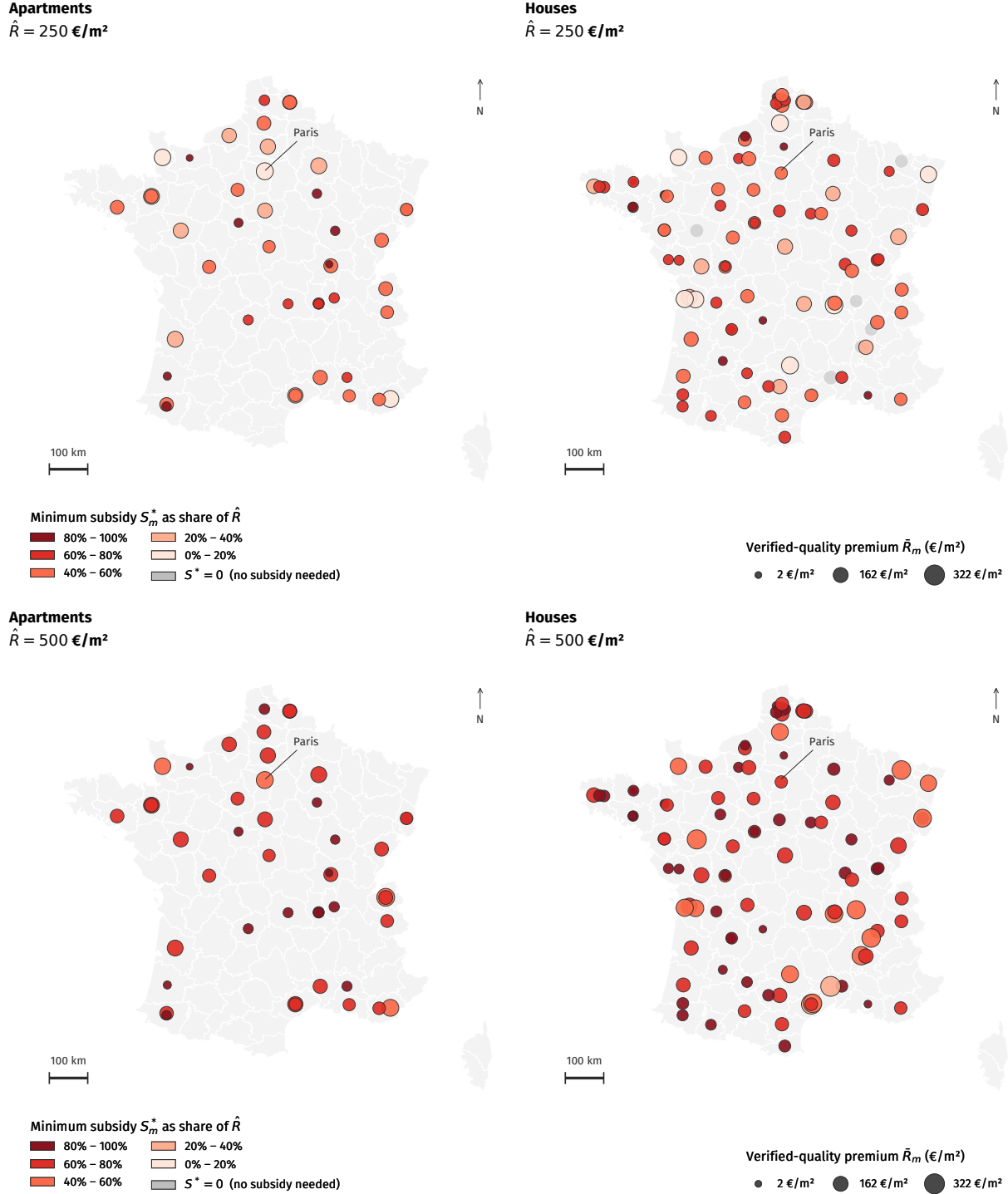


Notes. Each bar sums to 100% of markets in its (property type, $\hat{R}$) cell. The coloured stack marks markets requiring a positive subsidy; the grey residual labels markets with $S_m^* = 0$, for which the verified-quality premium already exceeds the renovation cost. Benchmarks shown: $\hat{R} \in \{250, 500\}$ €/m². Every market in the estimation sample is in the Partial Pooling regime, so there is no corner-regime subset to separate out; the threshold $\bar{R}_m$ is invariant to c_0 throughout (Lemma 5.1).

than idiosyncratic, since the cost-sharing ratio is mechanically low where $\bar{R}_m$ is high, and $\bar{R}_m$ is high where local prices per m² are large and the energy-efficiency premium $\beta_{1,m}$ itself is large. The latter reflects the capitalised value of verified quality to sensitive buyers, not a separate, freestanding measure of how many buyers are sensitive (Appendix B shows the two are not identified apart from one another in price data alone). These conditions hold in the high-price urban markets of metropolitan France, which carry the largest bubbles on the maps; that ratio is correspondingly higher in lower-price peripheral and rural markets, where small bubbles signal that the verified-quality premium delivers only a modest fraction of the renovation cost. An efficient programme would differentiate the per-m² subsidy along these structural axes rather than apply a uniform amount, channelling more public expenditure to markets where the private renovation gap is largest. The bubble-size dimension also points to the co-targeting logic of §5. Markets with small bubbles typically have low verification penetration π , combining a small per-renovation base with a small eligible population. Effective deployment therefore requires simultaneously raising π through audit-infrastructure investment and concentrating that transfer on the markets where both margins are most constrained.

Aggregating S_m^* over all listings yields upper-bound programme costs under the counterfactual that every eligible listing is renovated. For apartments (633,399 listings), the aggregate is €2.5 billion at $\hat{R} = 250$ and €11.9 billion at $\hat{R} = 500$. For houses (500,415 listings), the corresponding figures are €7.3 billion and €21.2 billion. The combined totals are €9.8 billion at the light benchmark and €33.0 billion at the medium benchmark. MaPrimeRénov's current annual budget is approximately €4–5 billion; a light-intervention rollout would therefore require roughly 2–2.5× current spending, rising to 7–8× at the medium benchmark. That multiplier is the headline fiscal fact. It implies that any credible decarbonisation programme targeting the housing stock must either substantially expand the fiscal envelope or rely heavily on the cost-sharing and co-targeting logic developed above to concentrate expenditure where the private-renovation gap is largest. These estimates should be read as order-of-magnitude

Figure 13. Spatial distribution of the minimum incentive-compatible subsidy across metropolitan France



Notes. Top row: $\hat{R} = 250 \text{ €/m}^2$ (light intervention). Bottom row: $\hat{R} = 500 \text{ €/m}^2$ (medium intervention). Left panels show apartment markets; right panels show house markets. Projection: Lambert-93 (EPSG:2154); the bar in the lower-left of each panel measures 100 km. Bubble colour encodes the cost-sharing ratio $S_m^*/\hat{R}$, the share of the renovation cost the regulator must cover; bins are fixed at $\{0\text{--}20\%, 20\text{--}40\%, 40\text{--}60\%, 60\text{--}80\%, 80\text{--}100\%\}$ so colours are directly comparable across cost benchmarks. Bubble size encodes the renovation threshold $\tilde{R}_m$ (verified-quality premium in euros per m²), pooled across both property types and both benchmarks so sizes are comparable across all four panels. Markets with $S_m^* = 0$ are displayed as small uniform grey dots.

upper bounds. The per-listing aggregation abstracts from the extensive-margin participation decision of sellers, and the realised fiscal cost under any actual programme would be smaller. More fundamentally, Remark 5.3 establishes that S_m^* is an incentive-compatibility object rather than a Ramsey-optimal transfer. With a shadow cost of public funds $\lambda > 1$, the regulator will set the operative subsidy below S_m^* , trading off the gap between the renovation cost and the verified-quality premium against the deadweight cost of taxation. The maps and distributions above identify where market failures are largest; they are properly read as identifying the targets of maximum social return per euro of public expenditure, not as direct programme prescriptions.

6 Conclusion

Four years after France mandated energy performance disclosure, silence is still rewarded. Properties listed without an EPC score trade at a premium of 6.9 percent above disclosed low-efficiency properties for apartments and 3.4 percent for houses. Standard unraveling theory predicts the opposite. This paper resolves the puzzle.

The resolution requires two departures from the classical framework, each empirically motivated and individually insufficient. First, not all sellers are informed. At listing time, only 51.8% of apartment sellers and 41.7% of house sellers hold a certified energy audit. When buyers observe a missing EPC score, they face genuine ambiguity. The non-disclosing seller may be concealing an unfavorable rating or may simply not know her property's energy performance. Uninformed sellers therefore provide strategic hiders with involuntary cover, preventing silence from being read as the worst possible signal. Second, not all buyers price energy performance into their offers. When the share of energy-sensitive buyers is moderate, the revenue penalty from concealment is diluted, making hiding profitable even against a skeptical pool. Neither departure alone is sufficient. Uninformed sellers keep silence ambiguous, but buyer heterogeneity ensures that exploiting this ambiguity is financially worthwhile.

These two forces generate three equilibrium regimes. In the partial-pooling equilibrium that describes the bulk of French housing markets, the fraction of informed low-quality sellers that hides adjusts endogenously until the marginal hider is exactly indifferent between concealment and disclosure. This indifference condition delivers the paper's central theoretical result. The non-disclosure premium equals exactly the concealment cost in partial pooling, invariant to the share of informed sellers and the share of energy-sensitive buyers. We identify the concealment cost directly from the disclosed-price gap and the registry-measured belief, without needing buyer energy-sensitivity to be separately estimated, and test the indifference prediction against it rather than assume it. The two segments concur less than an earlier reading of these facts suggested. Concealment costs are 2.4% of property value for apartments and 3.5% for houses, of the same order across two segments that differ in buyer energy-sensitivity (49.5% for apartments, 45.8% for houses, identified from an external energy-cost anchor rather than from the price data used to construct the premium itself); but the indifference prediction itself holds for houses and is decisively rejected for apartments, where the non-disclosure premium exceeds what the concealment cost and the registry-measured belief predict by a wide and precisely estimated margin. Strategic hiding is itself uncommon. 15.2% of informed apartment sellers with unfavorable ratings withhold them, versus 6.8% for houses. The non-disclosure premium survives not because strategic hiding is pervasive but because genuine uninformedness is. The uninformed seller, who pays no concealment cost yet receives the same premium as the strategic hider, is the residual claimant on the information rent; the informed hider, in the segment where the model fits, merely breaks even.

The policy implications depart from textbook prescriptions in one consequential way. Although the renovation threshold mechanically incorporates the concealment cost, the seller's indifference condition causes that cost to cancel exactly in partial pooling. The renovation incentive reduces to the verified-quality premium, the price gap between high-quality and low-quality disclosed properties, and is invariant to both the concealment cost and the share of informed sellers. Tightening enforcement against non-disclosure therefore does not move the renovation threshold in the empirically dominant regime. The regulator faces three distinct instruments: renovation subsidies (closing the gap between the verified-quality premium and the renovation cost), enforcement against strategic hiding (relevant only outside partial pooling), and audit-infrastructure investment (scaling how many sellers the other two instruments reach, not the per-seller incentive itself). These are complements, not substitutes. A uniform national subsidy exploits none of this structure. The minimum cost-sharing requirement ranges from zero in markets where the private incentive already suffices to the full renovation cost where the verified-quality premium is small, and aggregate programme costs reach approximately 9.8 billion euros at the light renovation benchmark, roughly two to two-and-a-half times the current annual budget of MaPrimeRénov', rising to approximately 33

billion euros and seven to eight times the current budget at the medium benchmark. Efficient targeting must condition on local market composition, directing public expenditure to markets where both the private renovation gap and the unverified seller population are largest.

The broader lesson extends beyond energy labels. Disclosure mandates discipline strategic silence only when sellers are actually informed. When a substantial share genuinely does not know their product's quality, the mandate produces ambiguity rather than transparency. Involuntary non-disclosers shelter strategic hidens, and silence cannot be read as bad news. The policy challenge is not to punish opacity more severely but to produce knowledge in the first place. In housing energy markets, as in any market where non-disclosure mixes strategy with ignorance, audit-infrastructure investment is the instrument that expands the reach of every other tool in the regulator's toolkit.

A Robustness: The Quality Threshold and the $\sigma_1^* = 0$ Restriction

Two maintained choices in the baseline specification deserve separate scrutiny, because each is an assumption rather than an estimate.

The first is where the binary quality space is cut. The baseline places the boundary between D and E, so that $\theta = 1$ corresponds to grades A–D. This follows the rental-ban discontinuity of *loi Climat et Résilience* discussed in Section 3.1, and it is the cut for which the price-index contrast and the listing-level quality measure refer to the same partition of grades. The natural alternative places the boundary between C and D.

The second is the treatment of informed sellers who withhold. Because such a listing displays no score, its quality cannot be read from the listing itself. The baseline assigns $\theta = 0$ by revealed preference, invoking Corollary 4.1. That assignment is an assumption, and its consequence is worth stating plainly. It makes $\hat{\sigma}_{1,m}$ identically zero by construction, so the corollary cannot be assessed against the baseline estimates. The alternative reads the withheld grade from the ADEME registry, which records it for every matched listing, leaving $\hat{\sigma}_{1,m}$ measured.

Table 6 crosses the two choices.

Table 6. Sensitivity to the quality threshold and to the σ_1 assignment

Specification	N	$\hat{\rho}_m$	$\hat{c}_{0,m}$	$\hat{\sigma}_{0,m}$	$\hat{\sigma}_{1,m}$	$\hat{\theta}_m$
<i>Panel A: Apartments</i>						
A–D vs E/F/G, σ_1 imposed (baseline)	45	49.5%	2.41%	15.2%	n/a	62.5%
A–D vs E/F/G, σ_1 from registry	45	49.5%	2.62%	5.9%	5.1%	65.9%
A–C vs E/F/G, σ_1 imposed	40	62.2%	1.08%	7.7%	n/a	26.5%
A–C vs E/F/G, σ_1 from registry	40	62.2%	1.19%	5.1%	6.5%	28.4%
<i>Panel B: Houses</i>						
A–D vs E/F/G, σ_1 imposed (baseline)	97	45.8%	3.45%	6.8%	n/a	60.5%
A–D vs E/F/G, σ_1 from registry	97	45.8%	3.56%	2.8%	2.1%	61.9%
A–C vs E/F/G, σ_1 imposed	100	53.7%	1.72%	3.7%	n/a	28.5%
A–C vs E/F/G, σ_1 from registry	100	53.7%	1.78%	2.5%	2.5%	29.3%

Notes. Market-level means. “ σ_1 imposed” assigns $\theta = 0$ to informed non-disclosers by revealed preference, which forces $\hat{\sigma}_{1,m} = 0$ in every market; the column reads n/a there because the quantity is assumed rather than estimated. “ σ_1 from registry” binarises the withheld grade from the ADEME record. The quality threshold also governs the price-index contrast. A–D vs E/F/G merges the *Efficient* and *Balanced* index series by transaction volume, while A–C vs E/F/G contrasts *Efficient* against *Intensive* and omits *Balanced*. Market counts differ across rows because the minimum-volume filter is applied to the high-group series, which is larger under the A–D contrast.

Three results follow.

First, buyer sensitivity is exactly insensitive to the σ_1 assignment, and the concealment cost only mildly so. Holding the threshold fixed, relaxing the restriction leaves $\hat{\rho}_m$ completely unchanged in both segments: the anchored estimator $\hat{\rho}_m = \hat{\beta}_{1,m}/\hat{\beta}_{1,m}^{\text{anchor}}$ (Proposition B.3) never involves σ_1 at all, so there is nothing for the assignment to move. $\hat{c}_{0,m}$ moves modestly, from 2.41% to 2.62% for apartments and from 3.45% to 3.56% for houses, through its dependence on $\hat{e}_m^*$, which does respond to the assignment. The concealment cost is therefore not an artefact of the revealed-preference assignment, even though $\hat{\sigma}_{0,m}$ itself is quite sensitive to it: imposing $\theta = 0$ places every informed non-discloser in the low-quality group and inflates the measured hiding rate by roughly a factor of two and a half (15.2% against 5.9% for apartments, 6.8% against 2.8% for houses).

Second, the restriction $\sigma_1^* = 0$ is not satisfied exactly, but the model’s predicted *ordering* holds under the maintained threshold. With quality read from the registry and the cut at the rental-ban boundary, $\hat{\sigma}_{1,m} = 5.1\%$ against $\hat{\sigma}_{0,m} = 5.9\%$ for apartments, and 2.1% against 2.8% for houses. Informed sellers of low-quality properties withhold more often than informed sellers of high-quality ones, as the model requires, though the gap is narrow. Under the C–D cut the ordering reverses for apartments (6.5% against 5.1%) and is effectively a tie for houses (2.5% against 2.5%, indistinguishable at this precision). The threshold that the institutional setting selects is

thus also the one under which the behavioural restriction survives, which we read as support for the baseline cut rather than as independent confirmation of the corollary.

Third, the threshold matters a great deal for the level of the concealment cost, and this is the paper's most consequential specification choice. Moving the cut from C–D to D–E raises $\hat{c}_{0,m}$ from roughly 1.1–1.8% of property value to 2.4–3.6%, a little more than a doubling in both segments. The mechanism is transparent. With A–D counted as high quality, the average quality of the non-disclosing pool $\hat{e}_m^*$ rises substantially (the informed high-quality share $\hat{\theta}_m$ alone more than doubles, from roughly 27–29% under the C–D cut to 60–66% under D–E), and $c_{0,m} = \beta_{1,m} \hat{e}_m^*$ (Proposition B.2) scales with it. Readers who prefer the narrower definition of “high quality” should therefore read the concealment cost as the smaller figure throughout; the relative ordering of the concealment cost across apartments and houses is unaffected by the threshold choice.

B Identification of ρ_m : A Formal Treatment

The naive market-by-market ratio $\hat{\rho}_m = \hat{\Delta}_m / (\hat{\beta}_{1,m} \hat{e}_m^*)$ is degenerate: it returns approximately 1 regardless of the true buyer-sensitivity share, because ρ_m cancels between numerator and denominator under the model's own pricing equations. This appendix states the identification failure formally, and the external anchor that resolves it.

The scale non-identification result

Write $b_m \equiv \rho_m \beta_1$ for the composite parameter that governs every observable price implication of the model.

Proposition B.1 (Scale non-identification of buyer sensitivity) *Under the benchmark pricing rule $\alpha_k = \beta_k$ (Section 4), fix any parameter vector with $\rho_m \in (0, 1]$ and $\beta_1 > 0$ for all markets m , and any $\lambda \in (0, (\max_m \rho_m)^{-1}]$. The reparametrization $(\rho_m, \beta_1) \mapsto (\lambda \rho_m, \beta_1 / \lambda)$ is admissible and induces an identical joint distribution of observed prices, an identical equilibrium hiding rate σ_0^* , and an identical equilibrium regime, in every market and every period. Consequently $\{\rho_m\}$ and β_1 are not separately identified by any price-based moment implied by the model, for any number of markets M ; only the composites $\{b_m\}$ and $\{c_{0,m}\}$ are.*

Proof B.1 Every observable implication of the model is a function of (ρ_m, β_1) through the product b_m alone. For a disclosed listing, $p^*(d = 1, \theta) = \beta_0 + \rho_m \beta_1 \theta = \beta_0 + b_m \theta$. For a non-disclosed listing, $p^*(d = 0) = \beta_0 + \rho_m \beta_1 e = \beta_0 + b_m e$. The hiding incentive is $\Delta(\sigma_0) = \rho_m \beta_1 e(\sigma_0) - c_{0,m} = b_m e(\sigma_0) - c_{0,m}$, so the equilibrium σ_0^* solving $\Delta(\sigma_0^*) = 0$ depends on (ρ_m, β_1) only through b_m . The regime boundaries of Proposition 4.2 ($c_{0,m}$ against $b_m \bar{\theta}_m$ and $b_m e(1)$) involve (ρ_m, β_1) the same way. Since $b_m = \rho_m \beta_1 = (\lambda \rho_m)(\beta_1 / \lambda)$ is invariant under the stated map and $c_{0,m}$ is untouched, the reparametrized vector reproduces every observable channel exactly. Admissibility requires $\lambda \rho_m \leq 1$ for all m , i.e. $\lambda \leq (\max_m \rho_m)^{-1}$.

Corollary B.1 (The raw ratio is a specification-test statistic, not an estimator) $\hat{\rho}_m = \hat{\Delta}_m / (\hat{\beta}_{1,m} \hat{e}_m^*)$ satisfies $\text{plim } \hat{\rho}_m = 1$ for every true $\rho_m \in (0, 1]$, since under the model $\hat{\Delta}_m = b_m e_m^*$ and $\hat{\beta}_{1,m} = b_m$ share the same composite b_m by Proposition B.1. It is the sample analogue of the over-identifying restriction $\Delta_m = \beta_{1,m} e_m^*$ (Lemma B.1 below), not an estimate of buyer sensitivity; its deviation from unity is a specification-test statistic, and clipping it to $[0, 1]$ suppresses that test rather than estimating anything.

What is identified without an external anchor

Proposition B.2 (Identification of the composite and of c_0) *Given consistent estimates of $(\pi_m, \bar{\theta}_m, \sigma_{0,m})$ from the ADEME registry match and $\hat{\beta}_{1,m}$ from the disclosed-price DiD:*

- (i) b_m is point identified: $b_m = \beta_{1,m}$.
- (ii) If $\sigma_{0,m} \in (0, 1)$ (Partial Pooling), $c_{0,m}$ is point identified, without reference to ρ_m or β_1 separately: $c_{0,m} = \beta_{1,m} \cdot e(\sigma_{0,m}; \pi_m, \bar{\theta}_m)$.
- (iii) If $\sigma_{0,m} = 0$ (Cost-Deterred Hiding), $c_{0,m} \geq \beta_{1,m} \bar{\theta}_m$; if $\sigma_{0,m} = 1$ (Complete Adverse Selection), $c_{0,m} \leq \beta_{1,m} e(1)$.
- (iv) Δ_m is an over-identifying restriction: $\Delta_m = \beta_{1,m} e(\sigma_{0,m})$, i.e. $\Delta_m = c_{0,m}$ in Partial Pooling.
- (v) ρ_m itself is not identified (Proposition B.1).

Proof B.2 (i) is the disclosed-price equation $\beta_{1,m} = p^*(d = 1, \theta = 1) - p^*(d = 1, \theta = 0) = b_m$. (ii): in Partial Pooling the indifference condition $\Delta(e(\sigma_0^*)) = c_{0,m}$ reads $b_m e(\sigma_{0,m}) = c_{0,m}$; substitute $b_m = \beta_{1,m}$ from (i). (iii) follows from the regime boundaries of Proposition 4.2 evaluated at $b_m = \beta_{1,m}$. (iv) combines (i) with the non-disclosed pricing equation $\Delta_m \equiv p^*(d = 0) - p^*(d = 1, \theta = 0) = b_m e(\sigma_{0,m})$. (v) is Proposition B.1.

This result has an immediate practical consequence: $\hat{c}_{0,m} = \hat{\beta}_{1,m} \cdot \hat{e}_m^*(\hat{\sigma}_{0,m})$ is the primary concealment-cost estimator used in Section 4.4, for both property types, and it does not require ρ_m to be identified at all.

Lemma B.1 (Model-implied bounds on the non-disclosure premium) (i) $0 \leq \Delta_m / \beta_{1,m} = e_m \leq \bar{\theta}_m \leq 1$, since the belief function is bounded above by $\bar{\theta}_m$ (attained at $\sigma_0 = 0$). (ii) In Partial Pooling, if $c_{0,m}$ is invariant to the July 2021 reform (the identifying assumption of Section 4.4), the model predicts a DiD non-disclosure premium of zero: since $\Delta_{m,t} = c_{0,m}$ in Partial Pooling at every date t , its post-minus-pre change is zero whenever $c_{0,m}$ does not move at the reform.

Point identification under an external anchor

Assumption B.1 (Energy-cost anchor) For each market m , an external, non-price estimate of the sensitive buyer's willingness to pay for verified quality is available as the capitalised energy-cost saving:

$$\beta_{1,m}^{\text{anchor}} = \kappa \cdot \frac{A_m}{P_m}, \quad A_m = (\bar{E}_{0,m} - \bar{E}_{1,m}) \cdot p^E \cdot a(r, T), \quad a(r, T) = \frac{1 - (1 + r)^{-T}}{r}, \quad (\text{Eq. 15})$$

where $\bar{E}_{\theta,m}$ is mean certified primary-energy consumption ($\text{kWh}_{\text{ep}}/\text{m}^2/\text{yr}$) of quality- θ dwellings among ADEME-matched ($K = 1$) listings in market m — pooling both disclosed and strategically-hidden listings, since restricting to disclosed-only listings would reintroduce selection on the seller's own hiding decision — p^E is a blended primary-energy-equivalent price, P_m is the local price level, and $a(r, T)$ discounts the annual saving over a horizon T at rate r .

Proposition B.3 (Point identification of ρ_m under Assumption B.1) Under Assumption B.1 with $\beta_{1,m}^{\text{anchor}} > 0$ known,

$$\rho_m = \frac{\beta_{1,m}}{\beta_{1,m}^{\text{anchor}}}, \quad c_{0,m} = \beta_{1,m} \cdot e(\sigma_{0,m}; \pi_m, \bar{\theta}_m) \quad (\text{Eq. 16})$$

are point identified. The model's constraint $\rho_m \leq 1$ becomes the testable inequality $\beta_{1,m} \leq \beta_{1,m}^{\text{anchor}}$, rather than an imposed clip.

Proof B.3 The Jacobian of $(\beta_{1,m}, \sigma_{0,m})$ with respect to $(\rho_m, c_{0,m})$ at a Partial Pooling equilibrium, holding $\beta_{1,m}^{\text{anchor}}$ fixed, is

$$\begin{pmatrix} \beta_{1,m}^{\text{anchor}} & 0 \\ k \beta_{1,m}^{\text{anchor}} & -k \rho_m \beta_{1,m}^{\text{anchor}} / c_{0,m} \end{pmatrix}, \quad k \equiv \frac{(1 - \pi_m) \bar{\theta}_m}{c_{0,m} \pi_m (1 - \bar{\theta}_m)},$$

with determinant $-k \rho_m (\beta_{1,m}^{\text{anchor}})^2 / c_{0,m} \neq 0$ whenever $\rho_m, c_{0,m} > 0$ and $\pi_m, \bar{\theta}_m \in (0, 1)$. Local identification follows; global identification follows from strict monotonicity of $\beta_{1,m} \mapsto \rho_m$ in (Eq. 16) and, given ρ_m , of $\sigma_{0,m} \mapsto c_{0,m}$ through the strictly decreasing belief function of Lemma 4.1.

Diagnostics reported alongside the anchored estimates (Table 4 and Appendix A). The measured wedge $\hat{\lambda}_m \equiv \hat{\Delta}_m - \hat{\beta}_{1,m} \hat{e}_m^*$ (Lemma B.1/Corollary B.1) is reported per market rather than absorbed into $\hat{c}_{0,m}$; the concentration statistic $\hat{\Lambda}' = M^{-1} \sum_m \hat{\rho}_m (1 - \hat{\rho}_m)$, restricted to markets with $\hat{\rho}_m \in [0, 1]$, gates whether the anchored $\hat{\rho}_m$ sample carries enough interior variation to support a covariate index rather than only a common estimate; and the count of markets violating $\hat{\rho}_m \leq 1$ is reported directly rather than clipped away.

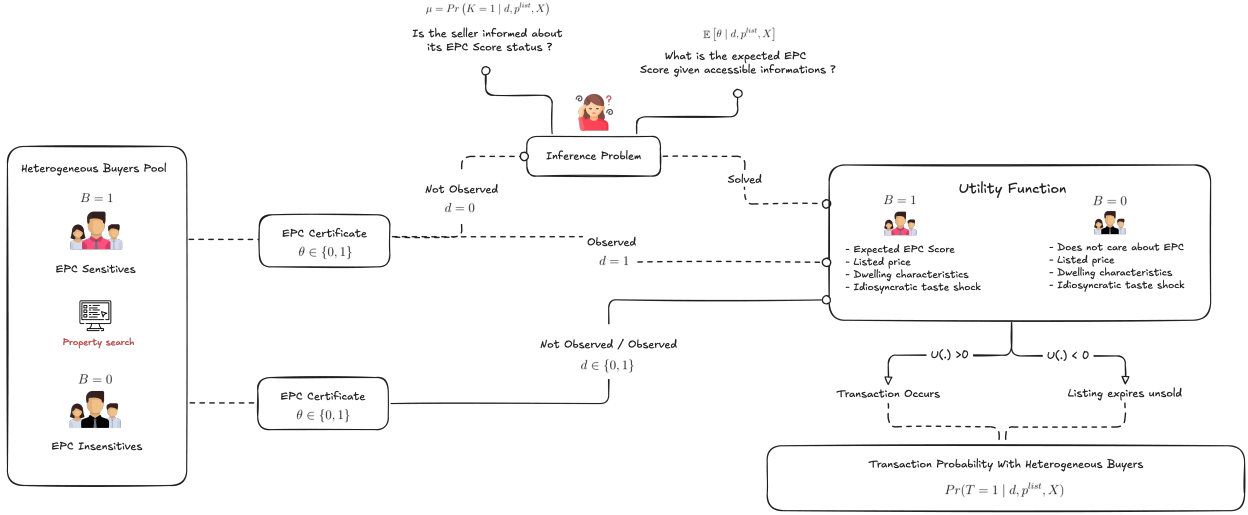
C Figures

D Data and Model Objects

Our dataset includes residential property listings from various French housing markets. For each listing i in market m , we observe:

- p : listing price

Figure 14. Visual representation of the model dynamics



- X : vector of property characteristics (such as size, age, location, etc.)
- $d \in \{0, 1\}$: indicator of whether the Energy Performance Certificate (EPC) score is displayed in the listing
- $EPC^{listing}$: the EPC score displayed on the listing (if $d_i = 1$)

What sets this approach apart is our ability to match these listings with data from ADEME (the French Environment and Energy Management Agency), which maintains an official registry of certified energy audits. These audits are reported by third-party assessors who are presumed to be honest and impartial. We match listings to the ADEME registry based on key characteristics that are registered in both databases, such as addresses, floor, item type, and area¹⁰. This allows us to assess the accuracy of the EPC scores displayed in listings by comparing them to the official certified energy audits. This allows us to observe notably:

- $K \in \{0, 1\}$: whether a verified EPC exists at listing time
- θ : the true (verified) EPC score (if $K = 1$)

This matching generates four mutually exclusive groups:

Table 7. Classification of Listings

Group	Displays EPC (d)	Verified EPC Exists (K)	Interpretation
Truthful Disclosure	1	1	Informed seller reveals θ
Strategic Hiding	0	1	Informed seller conceals θ
Uninformed	0	0	Seller lacks information
Falsification	1	0 or mismatch	Misreporting (excluded)

The key identification advantage is that we directly observe the latent variable K (information status), which the model treats as unobservable to buyers. This allows us to separately identify the behavior of informed and uninformed sellers.

¹⁰We provide a full explanation of our matching strategy in the appendix

E Proofs

All results below adopt the benchmark $\alpha_k = \beta_k$.

Preliminary: Optimal Price Sensitivity to Buyer Valuation

Proof E.1 (Optimal price sensitivity to shifts in buyer valuation) *The optimal price p^* satisfies the first-order condition:*

$$\frac{\gamma_1}{\gamma_0 + \gamma_1 \cdot p^*} = 1 - G(v_1 - p^*) \quad (\text{Eq. 17})$$

Define the left-hand side as $\text{LHS}(p) = \frac{\gamma_1}{\gamma_0 + \gamma_1 \cdot p}$ and the right-hand side as $\text{RHS}(p; v_1) = 1 - G(v_1 - p)$. The optimal price is determined by the intersection $\text{LHS}(p^*) = \text{RHS}(p^*; v_1)$.

Monotonicity of each side in p .

The left-hand side is strictly decreasing in p :

$$\frac{\partial \text{LHS}}{\partial p} = -\frac{\gamma_1^2}{(\gamma_0 + \gamma_1 \cdot p)^2} < 0 \quad (\text{Eq. 18})$$

The right-hand side is strictly increasing in p . Since G is the logistic CDF, $G'(x) = G(x)(1 - G(x)) > 0$ for all x . Therefore:

$$\frac{\partial \text{RHS}}{\partial p} = G'(v_1 - p) > 0 \quad (\text{Eq. 19})$$

A strictly decreasing function intersects a strictly increasing function at most once, so p^* is unique.

Effect of v_1 on the right-hand side.

At any fixed p , the right-hand side is strictly decreasing in v_1 :

$$\frac{\partial \text{RHS}}{\partial v_1} = -G'(v_1 - p) < 0 \quad (\text{Eq. 20})$$

An increase in v_1 shifts the RHS curve downward.

Comparative static.

Using the chain rule, differentiating the equilibrium condition with respect to v_1 :

$$\frac{\partial \text{LHS}}{\partial p} \cdot \frac{\partial p^*}{\partial v_1} = \frac{\partial \text{RHS}}{\partial p} \cdot \frac{\partial p^*}{\partial v_1} + \frac{\partial \text{RHS}}{\partial v_1} \quad (\text{Eq. 21})$$

Rearranging:

$$\frac{\partial p^*}{\partial v_1} = \frac{\frac{\partial \text{RHS}}{\partial v_1}}{\frac{\partial \text{LHS}}{\partial p} - \frac{\partial \text{RHS}}{\partial p}} = \frac{-G'(v_1 - p^*)}{-\frac{\gamma_1^2}{(\gamma_0 + \gamma_1 \cdot p^*)^2} - G'(v_1 - p^*)} > 0 \quad (\text{Eq. 22})$$

The numerator is negative. The denominator is negative (sum of two negative terms). Therefore $\frac{\partial p^*}{\partial v_1} > 0$.

Lemma A.1 (Profit is Increasing in Sensitive-Buyer Valuation)

Holding v_0 and ρ fixed, the gross profit function $\tilde{\Pi}(v_1; v_0, \rho)$ is strictly increasing in v_1 whenever $\rho > 0$.

Proof E.2 (Proof of Lemma A.1) Let $p_{\text{list}}^*(v_1)$ denote the optimal price given sensitive-buyer valuation v_1 . By the envelope theorem, the effect of a small increase in v_1 on maximized profit is:

$$\frac{\partial \tilde{\Pi}}{\partial v_1} = \phi(p_{\text{list}}^*) \cdot \rho \cdot G'(v_1 - p_{\text{list}}^*) \quad (\text{Eq. 23})$$

where we evaluate the partial derivative of the objective holding p fixed at the optimum.

All three terms are positive: $\phi(p^*) > 0$ (expected transaction price is positive), $\rho \in [0, 1] \geq 0$ (by assumption), and $G'(\cdot) > 0$ (logistic density is everywhere positive). Therefore $\frac{\partial \tilde{\Pi}}{\partial v_1} > 0$.

Since $\tilde{\Pi}$ is increasing in v_1 , the profit gain from disclosure is positive if and only if $v_1(d=1) > v_1(d=0)$: sensitive buyers value the property more when quality is revealed than when it is hidden. In this case, disclosure raises profit, and the seller discloses (absent hiding costs). Conversely, the profit gain is negative if $v_1(d=1) < v_1(d=0)$: sensitive buyers value the property more under hiding. This occurs when buyers' inference about non-disclosers is more favorable than the truth. In this case, hiding raises profit, and the seller hides (absent sufficiently high hiding costs).

Proof of Proposition 4.1 (Disclosure Incentives and Seller Sorting)

Proof E.3 By Lemma A.1, $p_0^* > p_1^*$ if and only if $v_1(d=0) > v_1(d=1, \theta)$. Since $v_1(d=0) - v_1(d=1, \theta) = \beta_1 \cdot (\mathbb{E}[\theta | d=0] - \theta)$ and $\beta_1 > 0$, this holds if and only if $\mathbb{E}[\theta | d=0] > \theta$.

- **For a low-quality seller** ($\theta = 0$): the condition is $\mathbb{E}[\theta | d=0] > 0$, which holds whenever the non-disclosing pool contains any high-quality sellers.
- **For a high-quality seller** ($\theta = 1$): the condition is $\mathbb{E}[\theta | d=0] > 1$, which is impossible since $\theta \in \{0, 1\}$.

Proof of Corollary 4.1 (High-Quality Sellers Always Disclose)

Proof E.4 From Proposition 4.1, the non-disclosure premium exists if and only if $\mathbb{E}[\theta | d=0] > \theta$.

For a high-quality seller ($\theta = 1$), this condition requires $\mathbb{E}[\theta | d=0] > 1$. But $\theta \in \{0, 1\}$, so $\mathbb{E}[\theta | d=0] \leq 1$ always. Therefore the condition never holds: high-quality sellers never earn a premium from hiding.

In fact, the inequality is strict whenever the non-disclosing pool contains any low-quality sellers. In that case $\mathbb{E}[\theta | d=0] < 1$, so:

$$v_1(d=1, \theta=1) - v_1(d=0) = \beta_1 \cdot (1 - \mathbb{E}[\theta | d=0]) > 0 \quad (\text{Eq. 24})$$

Disclosure yields a strictly higher valuation, and therefore strictly higher profit. Adding hiding costs $C(\theta=1) = c_0 + c_1 \geq 0$ only reinforces this: disclosure dominates hiding.

Proof of Proposition 4.2 (Equilibrium Existence and Uniqueness)

Proof E.5 By Corollary 4.1, $\sigma_1^* = 0$ in any equilibrium. The strategic problem reduces to characterising $\sigma_0^* \in [0, 1]$, the hiding probability of informed low-quality sellers. The incentive to hide is

$$\Phi(\sigma_0) \equiv \Delta(e(\sigma_0)) - c_0 = \frac{\rho \beta_1 (1 - \pi) \bar{\theta}}{\pi(1 - \bar{\theta})\sigma_0 + (1 - \pi)} - c_0,$$

where $e(\sigma_0)$ is the equilibrium pool belief (Equation (Eq. 5)). Since $e(\sigma_0)$ is strictly decreasing in σ_0 (Section 4.4.3) and Δ is strictly increasing in e , Φ is strictly decreasing in σ_0 , with boundary values

$$\Phi(0) = \Delta(\bar{\theta}) - c_0, \quad \Phi(1) = \Delta\left(\frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}\right) - c_0.$$

Case 1: Cost-Deterred Hiding. Suppose $c_0 > \Delta(\bar{\theta})$, i.e., $\Phi(0) < 0$.

Since Φ is strictly decreasing, $\Phi(\sigma_0) \leq \Phi(0) < 0$ for all $\sigma_0 \in [0, 1]$. Hiding is strictly unprofitable at every hiding rate, including the most favourable case $\sigma_0 = 0$. Disclosure strictly dominates for every informed low-quality seller, so the unique equilibrium is $\sigma_0^* = 0$ with equilibrium belief $e^* = \bar{\theta}$.

Interpretation. All informed low-quality sellers disclose; the non-disclosing pool consists exclusively of uninformed sellers. Buyers correctly infer quality $\bar{\theta}$ from non-disclosure. Unlike the classical Grossman–Milgrom outcome, silence is uninformative rather than damning: non-disclosure carries no stigma because it reveals nothing beyond the seller's lack of a certified audit.

Case 2: Complete Adverse Selection. Suppose $c_0 < \Delta(e(1))$, i.e., $\Phi(1) > 0$.

Since Φ is strictly decreasing, $\Phi(\sigma_0) \geq \Phi(1) > 0$ for all $\sigma_0 \in [0, 1]$. Hiding is strictly profitable at every hiding rate, including the least favourable case $\sigma_0 = 1$. Hiding is therefore strictly dominant for every informed low-quality seller regardless of the strategies of others, so the unique equilibrium is $\sigma_0^* = 1$ with equilibrium belief $e^* = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$.

Interpretation. All informed low-quality sellers pool with uninformed sellers. The non-disclosure premium $\rho \cdot \beta_1 \cdot e(1)$ is the maximum attainable across all regimes (Equation (Eq. 7)).

Case 3: Partial Pooling. Suppose $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$, i.e., $\Phi(1) \leq 0 \leq \Phi(0)$.

Φ is continuous and strictly decreasing on $[0, 1]$, with $\Phi(0) \geq 0$ and $\Phi(1) \leq 0$. By the intermediate value theorem, there exists $\sigma_0^* \in [0, 1]$ with $\Phi(\sigma_0^*) = 0$; strict monotonicity of Φ implies uniqueness. When the boundary inequalities hold strictly, σ_0^* lies in the open interval $(0, 1)$; at the boundaries, $\sigma_0^* \in \{0, 1\}$ and the regime collapses to Case 1 or Case 2.

At σ_0^* , informed low-quality sellers are exactly indifferent between hiding and disclosing; mixing at rate σ_0^* is self-confirming through the belief function. This regime generates the closed-form hiding rate and is the empirically dominant regime in the estimation sample.

Uniqueness and exhaustiveness. In Cases 1 and 2, uniqueness follows from strict dominance: Φ has constant sign on $[0, 1]$ so no other σ_0 is consistent with optimality. In Case 3, uniqueness follows from the unique zero of Φ . The three conditions $c_0 > \Delta(\bar{\theta})$, $c_0 < \Delta(e(1))$, and $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$ partition $\mathbb{R}_+$ since $\Delta(\bar{\theta}) \geq \Delta(e(1))$ (as $e(0) = \bar{\theta} \geq e(1)$ and Δ is increasing). Equilibrium exists and is unique for all parameter values.

Proof of Non-Disclosure Premium Characterisation (Eq. 7)

Proof E.6 Under the benchmark $\alpha_k = \beta_k$, the equilibrium price of a listing equals its expected buyer valuation. For a low-quality seller who discloses ($d = 1, \theta = 0$), both buyer types value the property at β_0 (sensitive buyers: $v_1(d = 1, \theta = 0) = \beta_0 + \beta_1 \cdot 0 = \beta_0$; insensitive buyers: $v_0 = \beta_0$), so $p_1^* = \beta_0$. For a non-disclosing listing, sensitive buyers assign expected quality $e(\sigma_0^*)$ and value the property at $\beta_0 + \beta_1 e(\sigma_0^*)$, while insensitive buyers value it at β_0 . Weighting by buyer composition yields $p_0^* = \beta_0 + \rho \beta_1 e(\sigma_0^*)$, so the non-disclosure premium is:

$$p_0^* - p_1^* = \rho \beta_1 e(\sigma_0^*). \quad (\text{Eq. 25})$$

The three cases of (Eq. 7) follow by substituting the equilibrium σ_0^* from each regime of Proposition 4.2 into (Eq. 25).

Case 1: Cost-Deterred Hiding ($\sigma_0^* = 0$). Evaluating (Eq. 5) at $\sigma_0 = 0$:

$$e(0) = \frac{(1 - \pi)\bar{\theta}}{(1 - \pi)} = \bar{\theta}.$$

Substituting into (Eq. 25): $p_0^* - p_1^* = \rho \beta_1 \bar{\theta}$.

Case 2: Complete Adverse Selection ($\sigma_0^* = 1$). With $\sigma_1^* = 0$ by Corollary 4.1, evaluating (Eq. 5) at $\sigma_0 = 1$:

$$e(1) = \frac{(1 - \pi)\bar{\theta}}{\pi(1 - \bar{\theta}) + (1 - \pi)} = \frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}}.$$

Substituting into (Eq. 25): $p_0^* - p_1^* = \rho \beta_1 \frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}}$.

Case 3: Partial Pooling ($\sigma_0^* \in (0, 1)$). By Proposition 4.2, σ_0^* solves the indifference condition $\Delta(e(\sigma_0^*)) = c_0$, i.e., $\rho \beta_1 e(\sigma_0^*) = c_0$. But the left-hand side is $p_0^* - p_1^*$ by (Eq. 25), so $p_0^* - p_1^* = c_0$.

Ordering. Since e is strictly decreasing by Lemma 4.1, we have $e(1) < e(\sigma_0^*) < e(0)$ for all $\sigma_0^* \in (0, 1)$, hence $\rho \beta_1 e(1) < c_0 < \rho \beta_1 \bar{\theta}$, confirming the strict ordering stated after (Eq. 7).

Proof of Proposition 4.3 (Comparative statics of the non-disclosure premium)

Proof E.7 From (Eq. 25), the non-disclosure premium is $\Delta \equiv p_0^* - p_1^* = \rho \beta_1 e(\sigma_0^*)$, where the belief $e(\sigma_0)$ is given by (Eq. 5). The equilibrium hiding rate $\sigma_0^* \in [0, 1]$ is determined by the regime characterised in Proposition 4.2. We derive $\partial\Delta/\partial\pi$ and $\partial\Delta/\partial\rho$ in each of the three regimes, in turn.

Case 1: Cost-Deterred Hiding ($\sigma_0^* = 0$). No informed low-quality seller hides, so the non-disclosing pool consists entirely of uninformed sellers. From (Eq. 5) at $\sigma_0 = 0$:

$$e(0) = \frac{(1-\pi)\bar{\theta}}{(1-\pi)} = \bar{\theta}, \quad (\text{Eq. 26})$$

which is independent of π . The premium is therefore:

$$\Delta = \rho \beta_1 \bar{\theta}. \quad (\text{Eq. 27})$$

Since $\bar{\theta}$ is a structural parameter, $\partial\Delta/\partial\pi = 0$ and $\partial\Delta/\partial\rho = \beta_1 \bar{\theta} > 0$.

Case 2: Complete Adverse Selection ($\sigma_0^* = 1$). All informed low-quality sellers hide. From (Eq. 5) at $\sigma_0 = 1$:

$$e(1) = \frac{(1-\pi)\bar{\theta}}{\pi(1-\bar{\theta}) + (1-\pi)} = \frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}, \quad (\text{Eq. 28})$$

so $\Delta = \rho \beta_1 e(1)$. Differentiating with respect to π by the quotient rule:

$$\frac{\partial e(1)}{\partial \pi} = \frac{-\bar{\theta}(1-\pi\bar{\theta}) - (1-\pi)\bar{\theta} \cdot (-\bar{\theta})}{(1-\pi\bar{\theta})^2} = \frac{\bar{\theta}(\bar{\theta}-1)}{(1-\pi\bar{\theta})^2} = \frac{-\bar{\theta}(1-\bar{\theta})}{(1-\pi\bar{\theta})^2} < 0, \quad (\text{Eq. 29})$$

hence $\partial\Delta/\partial\pi = \rho \beta_1 \cdot \partial e(1)/\partial\pi < 0$. Differentiating with respect to ρ (with $e(1)$ determined solely by π and $\bar{\theta}$): $\partial\Delta/\partial\rho = \beta_1 e(1) > 0$.

Case 3: Partial Pooling ($\sigma_0^* \in (0, 1)$). The interior equilibrium satisfies $c_0 = \rho \beta_1 e(\sigma_0^*)$, so the premium is pinned at the hiding cost:

$$\Delta = c_0. \quad (\text{Eq. 30})$$

Changes in π or ρ induce compensating adjustments in σ_0^* that restore the indifference condition, leaving the premium invariant: $\partial\Delta/\partial\pi = 0$ and $\partial\Delta/\partial\rho = 0$.

To verify: from (Eq. 5), $\partial e(\sigma_0)/\partial\pi = -\bar{\theta}(1-\bar{\theta})\sigma_0/[\pi(1-\bar{\theta})\sigma_0 + (1-\pi)]^2 < 0$ for $\sigma_0 > 0$. Total differentiation of $c_0 = \rho \beta_1 e(\sigma_0^*(\pi), \pi)$ with respect to π gives:

$$0 = \rho \beta_1 \left(\frac{\partial e}{\partial \sigma_0} \frac{d\sigma_0^*}{d\pi} + \frac{\partial e}{\partial \pi} \right), \quad (\text{Eq. 31})$$

so $d\sigma_0^*/d\pi = -(\partial e/\partial\pi)/(\partial e/\partial\sigma_0) < 0$: as π rises, informed sellers populate the non-disclosing pool more heavily, so fewer low-quality sellers need to hide to sustain the same buyer belief, and σ_0^* falls.

Summary. The non-disclosure premium is weakly decreasing in π and weakly increasing in ρ , with strict monotonicity only in the Complete Adverse Selection regime. In Case 1, the premium is determined by the population quality $\bar{\theta}$ of the uninformed pool and is insensitive to market fundamentals π and ρ . In Case 3, the premium is pinned at the primitive c_0 by the indifference condition; it is the hiding rate σ_0^* , not the premium, that absorbs variation in π and ρ .

Proof of Corollary 4.2 (Effect of ρ on the non-disclosure premium)

Proof E.8 Part (i). From (Eq. 25), the non-disclosure premium is $\Delta = \rho \beta_1 e(\sigma_0^*)$. We differentiate with respect to ρ in each regime, holding the regime-specific equilibrium σ_0^* fixed where appropriate.

Cost-Deterred Hiding ($\sigma_0^* = 0$). Here $e(0) = \bar{\theta}$ by (Eq. 5), independent of ρ . Therefore $\partial\Delta/\partial\rho = \beta_1 \bar{\theta} > 0$.

Complete Adverse Selection ($\sigma_0^* = 1$). Here $e(1) = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$, also independent of ρ . Therefore $\partial\Delta/\partial\rho = \beta_1 e(1) > 0$.

Partial Pooling ($\sigma_0^* \in (0, 1)$). The indifference condition $\rho \beta_1 e(\sigma_0^*) = c_0$ pins $\Delta = c_0$. Changes in ρ induce compensating adjustments in σ_0^* , as established in Proposition 4.3(ii): $\partial\sigma_0^*/\partial\rho > 0$. The premium satisfies $\partial\Delta/\partial\rho = 0$.

Part (ii). From Proposition 4.2, the Partial Pooling regime requires $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$, i.e., $\rho \beta_1 e(1) \leq c_0 \leq \rho \beta_1 \bar{\theta}$. Differentiating both bounds with respect to ρ : $\partial[\rho \beta_1 e(1)]/\partial\rho = \beta_1 e(1) > 0$ and $\partial[\rho \beta_1 \bar{\theta}]/\partial\rho = \beta_1 \bar{\theta} > 0$. Both boundaries increase with ρ . The width of the partial-pooling interval is $\rho \beta_1 [\bar{\theta} - e(1)]$, which is increasing in ρ since $\bar{\theta} > e(1)$ whenever $\pi > 0$. The regime is therefore weakly expanding in ρ .

Proof of Proposition 4.4 (Equilibrium effect of π : equilibrium moves toward full unraveling)

Proof E.9 From Eq. (Eq. 5), the equilibrium belief under partial separation is:

$$e(1) = \frac{(1 - \pi)\bar{\theta}}{1 - \pi\bar{\theta}} \quad (\text{Eq. 32})$$

Taking the derivative:

$$\frac{\partial e(1)}{\partial \pi} = \frac{-\bar{\theta}(1 - \pi\bar{\theta}) - (1 - \pi)\bar{\theta}(-\bar{\theta})}{(1 - \pi\bar{\theta})^2} = \frac{-\bar{\theta}(1 - \bar{\theta})}{(1 - \pi\bar{\theta})^2} < 0 \quad (\text{Eq. 33})$$

As π increases, $e(1)$ falls: the pool quality under partial separation deteriorates because fewer uninformed sellers are available to provide cover.

Since $\Delta(e)$ is increasing in e , $\Delta(e(1))$ falls. The condition for partial separation, $c_0 < \Delta(e(1))$, becomes harder to satisfy. For fixed c_0 , an increase in π can push the equilibrium from partial separation into the mixed or full unraveling region.

When most sellers are informed (π high), non-disclosure is suspicious. Buyers recognize that uninformed sellers are rare, so most non-disclosers must be strategic hidrs. The favorable cover provided by uninformed sellers evaporates. Low-quality sellers find hiding less attractive, and the market approaches classical unraveling.

When most buyers care about quality (ρ high), the stakes of disclosure are high. Low-quality sellers have much to lose from revealing their type. The non-disclosure premium is large, making hiding attractive even when pool quality is low. Partial separation is sustained, and by Corollary 4.2(ii), the partial-pooling interval itself widens with ρ .

Proof of Proposition 4.5 (Equilibrium effect of c_0 : equilibrium moves toward full unraveling)

Proof E.10 An increase in c_0 shifts the function $\Phi(\sigma_0) = \Delta(e(\sigma_0)) - c_0$ downward. The boundaries become:

$$\begin{aligned} \Phi(0) &= \Delta(\bar{\theta}) - c_0 \quad (\text{decreases}) \\ \Phi(1) &= \Delta(e(1)) - c_0 \quad (\text{decreases}) \end{aligned} \quad (\text{Eq. 34})$$

The condition for full unraveling, $c_0 > \Delta(\bar{\theta})$, becomes easier to satisfy. The condition for partial separation, $c_0 < \Delta(e(1))$, becomes harder to satisfy.

Higher hiding costs make concealment less attractive. Regulatory enforcement, administrative burden, or reputational concerns that raise c_0 push the market toward disclosure.

Proof of Lemma 5.1 (Equilibrium reduction in partial pooling)

Proof E.11 Write $\Pi(d = 0) \equiv \Pi(d = 0; e^*)$ where the equilibrium belief is $e^* = e(\sigma_0^*)$ from (Eq. 5). The revenue $\Pi(d = 1, \theta)$ is independent of σ_0^* since buyers observe θ directly under disclosure (Corollary 4.1).

Partial Pooling. By Proposition 4.2, in the interior regime $\sigma_0^* \in (0, 1)$ is determined by the indifference condition

$$\Pi(d = 0; e^*) - c_0 = \Pi(d = 1, \theta = 0), \quad (\text{Eq. 35})$$

which states that an informed low-quality seller is indifferent between hiding (paying c_0 for the inflated pool revenue) and disclosing low quality. Substituting (Eq. 35) into the threshold definition (Eq. 11),

$$\bar{R}^{PP} = \Pi(d = 1, \theta = 1) - \Pi(d = 0; e^*) + c_0 = \Pi(d = 1, \theta = 1) - \Pi(d = 1, \theta = 0).$$

The cancellation is exact: the apparent $+c_0$ in the threshold formula is offset by the equilibrium movement of $\Pi(d = 0; e^*)$ (which in PP exceeds $\Pi(d = 1, \theta = 0)$ by exactly c_0). $\bar{R}^{PP}$ depends on ρ and X only.

Complete Adverse Selection. The regime $\sigma_0^* = 1$ obtains because hiding strictly dominates disclose-low at the equilibrium pool composition $e^{CAS} = (1 - \pi)\bar{\theta}/(1 - \pi\bar{\theta})$: $\Pi(d = 0; e^{CAS}) - c_0 > \Pi(d = 1, \theta = 0)$. The corner is locally stable in c_0 within the regime's defining inequality, so σ_0^* and e^* are locally constant in c_0 , and the threshold retains its primitive form $\bar{R}^{CAS} = \Pi(d = 1, \theta = 1) - \Pi(d = 0; e^{CAS}) + c_0$ with $\partial \bar{R}^{CAS}/\partial c_0 = 1$.

Proof of Proposition 5.1 (Subsidy comparative statics by regime)

Throughout, we use the regime-conditional form of $\bar{R}$ from Lemma 5.1. We assume G is log-concave so that $\Pi(d, \theta; e) = \max_p \phi(p)\Gamma(p \mid d, \theta; e)$ admits a unique interior maximiser $p^*(d, \theta; e)$ and is differentiable in e . The optimal subsidy $S^* = \max\{\hat{R}(X) - \bar{R}, 0\}$ is piecewise linear in its arguments; we evaluate the derivatives on the active region $\hat{R}(X) > \bar{R}$. The relation $\partial S^*/\partial \bar{R} = -1$ then translates each derivative of $\bar{R}$ into the corresponding derivative of S^* .

Partial Pooling

Proof E.12 (Effect of c_0) From Lemma 5.1, in PP, $\bar{R}^{PP} = \Pi(d = 1, \theta = 1) - \Pi(d = 1, \theta = 0)$. Both terms are verified-disclosure revenues with θ observable; neither depends on the equilibrium hiding rate, the non-disclosing pool, or the cost of hiding. Hence $\partial \bar{R}^{PP}/\partial c_0 = 0$, and consequently $\partial S^*/\partial c_0 = 0$ when $\hat{R}(X) > \bar{R}^{PP}$.

Mechanism. The equilibrium indifference (Eq. 35) forces $\Pi(d = 0; e^*)$ to track c_0 one-for-one through the equilibrium response of σ_0^* , so the apparent $+c_0$ in the primitive threshold formula (Eq. 11) is exactly cancelled by the equilibrium movement of $\Pi(d = 0; e^*)$. The renovation incentive reduces to the verified-quality premium, which involves no hiding-related object.

Proof E.13 (Effect of π) The verified-quality revenues $\Pi(d = 1, \theta \in \{0, 1\})$ are independent of the non-disclosing pool composition and hence of π . Therefore $\partial \bar{R}^{PP}/\partial \pi = 0$ and $\partial S^*/\partial \pi = 0$.

Proof E.14 (Effect of ρ) Buyer valuations under disclosure satisfy $v_1(d = 1, \theta) = \beta_0 + \beta_1\theta + \sum_n \beta_n X_n$ and $v_0 = \beta_0 + \sum_n \beta_n X_n$. The aggregate demand under disclosure is

$$\Gamma(p \mid d = 1, \theta) = \rho G(v_1(d = 1, \theta) - p) + (1 - \rho) G(v_0 - p).$$

By the envelope theorem applied to $\Pi(d = 1, \theta) = \max_p \phi(p)\Gamma(p \mid d = 1, \theta)$,

$$\frac{\partial \Pi(d = 1, \theta)}{\partial \rho} = \phi(p^*) [G(v_1(d = 1, \theta) - p^*) - G(v_0 - p^*)].$$

Subtracting the $\theta = 0$ from the $\theta = 1$ case (noting $v_1(d = 1, \theta = 0) = v_0$, so the bracket is zero at $\theta = 0$ and strictly positive at $\theta = 1$ since $\beta_1 > 0$ and G is strictly increasing):

$$\frac{\partial \bar{R}}{\partial \rho} = \frac{\partial \Pi(d = 1, \theta = 1)}{\partial \rho} - \frac{\partial \Pi(d = 1, \theta = 0)}{\partial \rho} > 0,$$

where the optimal price p^* in each term is evaluated at its own argument. Consequently $\partial S^*/\partial \rho \leq 0$ in the active region.

Complete Adverse Selection

Proof E.15 (Effect of c_0) In CAS, $\sigma_0^* = 1$ is locally constant in c_0 within the regime's defining inequality, hence e^{CAS} is independent of c_0 and so is $\Pi(d = 0; e^{CAS})$. Differentiating $\bar{R}^{CAS} = \Pi(d = 1, \theta = 1) - \Pi(d = 0; e^{CAS}) + c_0$ directly yields $\partial \bar{R} / \partial c_0 = 1$, hence $\partial S^* / \partial c_0 = -1$ when $\hat{R}(X) > \bar{R}$. The textbook substitutability between hiding-cost enforcement and renovation subsidies thus holds only at this corner.

Proof E.16 (Effect of π) In CAS, $e^{CAS} = (1 - \pi)\bar{\theta} / (1 - \pi\bar{\theta})$ with $\partial e^{CAS} / \partial \pi < 0$. By the envelope theorem,

$$\frac{\partial \Pi(d = 0; e^{CAS})}{\partial \pi} = \frac{\partial \Pi(d = 0; e^{CAS})}{\partial e} \cdot \frac{\partial e^{CAS}}{\partial \pi},$$

with $\partial \Pi / \partial e = \phi(p^*)\rho\beta_1 g(v_1(d = 0) - p^*) > 0$ (because $v_1(d = 0) = \beta_0 + \beta_1 e + \sum_n \beta_n X_n$ is increasing in e and G is strictly increasing). Hence $\partial \Pi(d = 0; e^{CAS}) / \partial \pi < 0$ and $\partial \bar{R} / \partial \pi = -\partial \Pi(d = 0; e^{CAS}) / \partial \pi > 0$, giving $\partial S^* / \partial \pi \leq 0$.

Proof E.17 (Effect of ρ) The argument follows the PP case, with the additional step that $\Pi(d = 0; e^{CAS})$ depends on ρ through the share of EPC-sensitive buyers in aggregate demand; e^{CAS} itself is independent of ρ . Applying the envelope theorem to both $\Pi(d = 1, \theta = 1)$ and $\Pi(d = 0; e^{CAS})$ and using $v_1(d = 1, \theta = 1) > v_1(d = 0)$ yields $\partial \bar{R}^{CAS} / \partial \rho > 0$.

Effect of renovation cost $\hat{R}(X)$

Proof E.18 In the active region $\hat{R}(X) > \bar{R}$, the unconstrained form $S^* = \hat{R}(X) - \bar{R}$ applies, giving $\partial S^* / \partial \hat{R}(X) = 1$. In the inactive region, $S^* = 0$ and the derivative is zero. The indicator-function statement in Proposition 5.1 captures both.

Summary of policy reading. The substitutability $\partial S^* / \partial c_0 = -1$ obtains only in the Complete Adverse Selection corner, where the equilibrium hiding rate is at the boundary and $\Pi(d = 0; e^*)$ is locally constant in c_0 . In the partial-pooling regime, which describes the bulk of French apartment and house markets, the equilibrium indifference identity forces $\Pi(d = 0; e^*)$ to move one-for-one with c_0 , so the apparent $+c_0$ in the threshold formula is cancelled exactly and $\partial S^* / \partial c_0 = 0$. Disclosure enforcement and renovation subsidies are therefore policy complements rather than substitutes outside the CAS corner: enforcement shifts the within-stock composition of hidiers and disclosers along the indifference curve, while subsidies shift the stock itself toward verified high quality.

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